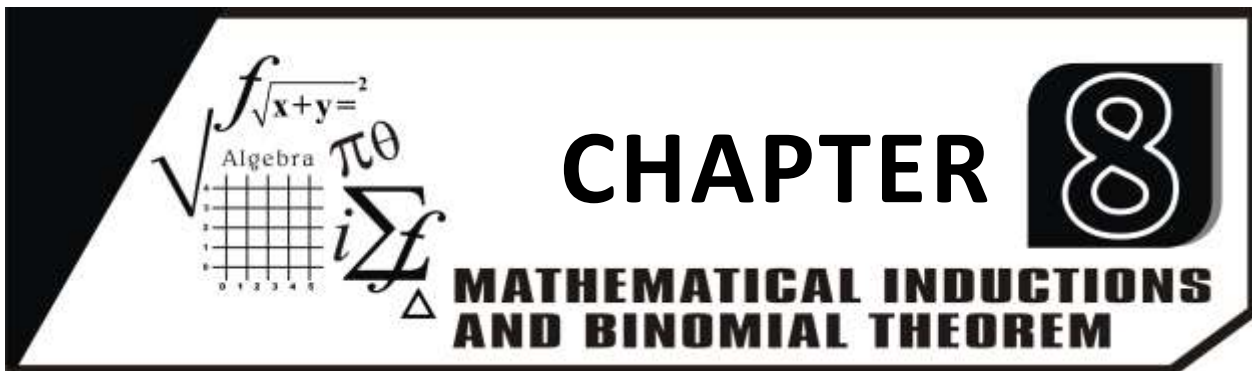




# CHAPTER 8

## MATHEMATICAL INDUCTIONS AND BINOMIAL THEOREM

### Introduction:

**Francesco Mourollico** (1494-1575) devised the method of induction and applied this device first to prove that the **sum of the first  $n$  odd positive integers equals  $n^2$** .

We are aware of the fact that even one exception or case to a mathematical formula is enough to prove it to be false. Such a case or exception which fails the mathematical formula or statement is called a **counter example**.

For example, we consider the statement  $S(n) = n^2 - n + 41$  is a prime number for every natural number  $n$ . The values of the expression  $n^2 - n + 41$  for some first natural numbers are given in the table as shown below.

|        |    |    |    |    |    |    |    |    |     |     |     |
|--------|----|----|----|----|----|----|----|----|-----|-----|-----|
| $n$    | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9   | 10  | 11  |
| $S(n)$ | 41 | 43 | 47 | 53 | 61 | 71 | 83 | 97 | 113 | 131 | 151 |

From the table, it appears that the statement  $S(n)$  has enough chance of being true. If we go on trying for the next natural numbers. We find  $n = 41$  as a counter example which fails the claim of the above statement. So we conclude that to derive a general formula without proof from some special cases is not a wise step. This example was discovered by **Euler** (1707-1783)

### Principle of Mathematical Induction

The principle of mathematical induction is stated as follows:

If a proposition or statement  $S(n)$  for each positive integer  $n$  is such that

1.  $S(1)$  is true i.e.,  $S(n)$  is true for  $n = 1$ .
2.  $S(k+1)$  is true whenever  $S(k)$  is true for any positive integer  $k$ ,

Then  $S(n)$  is true for all positive integers.

### Procedure:

1. Substituting  $n = 1$ , show that the statement is true for  $n = 1$ .
2. Assuming that the statement is true for any integer  $k$ , then show that it is true for the next higher integer.

M1: Starting with one side of  $S(k+1)$ , its other side is derived by using  $S(k)$ .

M2:  $S(k+1)$  is established by performing algebraic operations on  $S(k)$ .

### Principle of Extended Mathematical Induction:

Let  $i$  be an integer. If a formula or statement for  $n \geq i$  is such that

1.  $S(i)$  is true and
2.  $S(k+1)$  is true whenever  $S(k)$  is true for integral values of  $n \geq i$ .

**EXERCISE 8.1**

Use the mathematical induction to prove the following formulae for every positive integer  $n$ . (RWP 2021, LHR 2022)

**Q.1**  $1 + 5 + 9 + \dots + (4n - 3) = n(2n - 1)$

**Solution:**

Let  $S(n)$  be the give statement , i.e.,

$$S(n): 1 + 5 + 9 + \dots + (4n - 3) = n(2n - 1) \quad (i)$$

(i) when  $n = 1$ , Equation (i) becomes;

$$S(1): 4(1) - 3 = 1(2 \times 1 - 1)$$

$$S(1): 1 = 1$$

Thus  $S(1)$  is true i.e., condition (I) is satisfied.

(ii) Let us assume that  $S(n)$  is true for any  $n = k \in N$  i.e.,

$$1 + 5 + 9 + \dots + (4k - 3) = k(2k - 1) \quad (A)$$

The statement for  $n = k + 1$  becomes;

$$1 + 5 + 9 + \dots + (4k - 3) + (4k + 1) = (k + 1)(2k + 1) \quad (B)$$

Adding  $(4k + 1)$  on both sides of (A) we get;

$$\begin{array}{l|l} 1 + 5 + 9 + \dots + (4k - 3) + (4k + 1) = k(2k - 1) + (4k + 1) & \\ \hline = 2k^2 - k + 4k + 1 & = (2k + 1)(k + 1) \\ = 2k^2 + 3k + 1 & = (2k + 2 - 1)(k + 1) \\ = 2k^2 + 2k + k + 1 & = [2(k + 1) - 1](k + 1) \\ = 2k(k + 1) + 1(k + 1) & \end{array}$$

Thus  $S(k + 1)$  is true if  $S(k)$  is true, so condition (II) is satisfied. Since both the condition are satisfied, therefore,  $S(n)$  is true for all  $n \in N$ .

**Q.2**  $1 + 3 + 5 + \dots + (2n - 1) = n^2$

(LHR 2022)

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1 + 3 + 5 + \dots + (2n - 1) = n^2 \quad (i)$$

(i) when  $n = 1$ , equation (i) becomes;

$$S(1): 2 \times 1 - 1 = 1^2$$

$$S(1): 1 = 1$$

Thus  $S(1)$  is true that is condition (I) satisfied.

(ii) Let us assume that  $S(n)$  is true for any  $n = k \in N$ , i.e.,

$$S(k): 1 + 3 + 5 + \dots + (2k - 1) = k^2 \quad (A)$$

The statement for  $n = k + 1$  becomes ;

$$1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) = (k + 1)^2 \quad (B)$$

Adding  $(2k+1)$  on both sides of (A) we get;

$$\begin{aligned} 1+3+5+\dots+(2k-1)+(2k+1) &= k^2+(2k+1) \\ &= k^2+2k+1 \\ &= (k+1)^2 \end{aligned}$$

Thus  $S(k+1)$  is true if  $S(k)$  is true. So condition (II) is satisfied.

Since both the conditions are satisfied, therefore,  $S(n)$  is true for each positive integer  $n$ .

**Q.3**  $1+4+7+\dots+(3n-2) = \frac{n(3n-1)}{2}$

(RWP 2022, MTN 2023)

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1+4+7+\dots+(3n-2) = \frac{n(3n-1)}{2} \quad (i)$$

(i) When  $n=1$ ,  $S$  equation (i) becomes;

$$S(1): 3(1)-2 = \frac{1(3(1)-1)}{2}$$

$$S(1): 1=1$$

Thus  $S(1)$  is true, i.e., condition (I) is satisfied

(ii) Let us assume that  $S(n)$  is true for any  $n=k \in N$ , i.e.,

$$S(k): 1+4+7+\dots+(3k-2) = \frac{k(3k-1)}{2} \quad (A)$$

The statement for  $n=k+1$  becomes;

$$1+4+7+\dots+(3k-2)+(3k+1) = \frac{(k+1)(3k+2)}{2} \quad (B)$$

Adding  $(3k+1)$  on both sides of equation (A) we get

$$\begin{aligned} S(k): 1+4+7+\dots+(3k-2)+(3k+1) &= \frac{k(3k-1)}{2} + 3k+1 \\ &= \frac{3k^2 - k + 6k + 2}{2} &= \frac{(k+1)(3k+2+3-3)}{2} \\ &= \frac{3k^2 + 5k + 2}{2} &= \frac{(k+1)(3(k+1)-1)}{2} \\ &= \frac{3k^2 + 3k + 2k + 2}{2} \\ &= \frac{3k(k+1) + 2(k+1)}{2} \end{aligned}$$

Hence  $S(k+1)$  is true whenever  $S(k)$  is true so condition (II) is satisfied.

Since both the conditions are satisfied, therefore  $S(n)$  is true for each positive integer  $n$ .

**Q.4**  $1+2+4+\dots+2^{n-1}=2^n-1$  (FSD 2021, MTN 2023, LHR 2023)

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1+2+4+\dots+2^{n-1}=2^n-1 \quad (i)$$

(i) when  $n=1$ , equation (i) becomes;

$$S(1): 2^{1-1}=2^1-1$$

$$S(1): 2^0=2-1$$

$$S(1): 1=1$$

Thus  $S(1)$  is true that is condition (I) is satisfied.

(ii) Let us assume that  $S(n)$  is true for any  $n=k \in N$ , i.e.,

$$S(k): 1+2+4+\dots+2^{k-1}=2^k-1 \quad (A)$$

The statement for  $n=k+1$  becomes;

$$1+2+4+\dots+2^{(k+1)-1}=2^{k+1}-1 \quad (B)$$

Adding  $2^{(k+1)-1}$  on both sides of (A) we get;

$$\begin{array}{lcl} 1+2+4+\dots+2^{k-1}+2^{(k+1)-1} & = & (2^k-1)+2^{(k+1)-1} \\ = 2^k-1+2^k & & = 2^{k+1}-1 \\ = 2 \cdot 2^k-1 & & = 2^{(k+1)}-1 \end{array}$$

Thus  $S(k+1)$  is true if  $S(k)$  is true, so the condition (II) is satisfied.

Since both the conditions are satisfied, therefore,  $S(n)$  is true for each positive integer  $n$ .

**Q.5**  $1+\frac{1}{2}+\frac{1}{4}+\dots+\frac{1}{2^{n-1}}=2\left(1-\frac{1}{2^n}\right)$  (FSD 2022, GRW 2023)

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1+\frac{1}{2}+\frac{1}{4}+\dots+\frac{1}{2^{n-1}}=2\left(1-\frac{1}{2^n}\right) \quad (i)$$

(i) when  $n=1$ , equation (i) becomes;

$$S(1): \frac{1}{2^{1-1}}=2\left(1-\frac{1}{2^1}\right)$$

$$S(1): \frac{1}{2^0}=2\left(\frac{1}{2}\right)$$

$$S(1): 1=1$$

Thus  $S(1)$  is true that is condition (I) is satisfied.

(ii) Let us assume that  $S(k)$  is true for any  $n=k \in N$  that is

$$S(k): 1+\frac{1}{2}+\frac{1}{4}+\dots+\frac{1}{2^{k-1}}=2\left(1-\frac{1}{2^k}\right) \quad (A)$$

The given statement for  $n=k+1$  becomes;

$$1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{k-1}} + \frac{1}{2^k} = 2 \left( 1 - \frac{1}{2^{k+1}} \right) \quad (B)$$

Adding  $\frac{1}{2^k}$  on both sides of (A) we get;

$$\begin{aligned} 1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{k-1}} + \frac{1}{2^k} &= 2 \left( 1 - \frac{1}{2^k} \right) + \frac{1}{2^k} \\ &= 2 - \frac{2}{2^k} + \frac{1}{2^k} \\ &= 2 - \frac{1}{2^k} \end{aligned} \quad \left| \quad \begin{aligned} &= 2 \left( 1 - \frac{1}{2 \cdot 2^k} \right) \\ &= 2 \left( 1 - \frac{1}{2^{k+1}} \right) \end{aligned} \right.$$

Hence  $S(k+1)$  is true whenever  $S(k)$  is true.

So condition (II) is satisfied.

Since both conditions are satisfied therefore  $S(n)$  is true for each positive integer  $n$ .

**Q.6**  $2 + 4 + 6 + \dots + 2n = n(n+1)$  (GRW 2022, MTN 2023)

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 2 + 4 + 6 + \dots + 2n = n(n+1) \quad (i)$$

when  $n=1$ , equation (i) becomes;

$$(i) \quad S(1): 2(1) = 1(1+1)$$

$$S(1): 2 = 2$$

Thus  $S(1)$  is true that is condition (I) is satisfied.

(ii) Let us assume that  $S(n)$  is true for any  $n = k \in N$ , i.e.,

$$S(k): 2 + 4 + 6 + \dots + 2k = k(k+1) \quad (A)$$

The given statement for  $n = k+1$  becomes

$$2 + 4 + 6 + \dots + 2k + 2(k+1) = (k+1)(k+2) \quad (B)$$

Adding  $2(k+1)$  on both sides of (A) we get;

$$\begin{aligned} 2 + 4 + 6 + \dots + 2k + 2(k+1) &= k(k+1) + 2(k+1) \\ &= (k+1)(k+2) \end{aligned}$$

Hence  $S(k+1)$  true whenever  $S(k)$  is true. So condition (II) is satisfied.

Since both conditions are satisfied, therefore  $S(n)$  is true  $\forall n \in N$ .

**Q.7**  $2 + 6 + 18 + \dots + 2 \times 3^{n-1} = 3^n - 1$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 2 + 6 + 18 + \dots + 2 \times 3^{n-1} = 3^n - 1 \quad (i)$$

(i) when  $n=1$ ,  $S(1)$  becomes;

$$S(1): 2 \times 3^{1-1} = 3^1 - 1$$

$$S(1): 2 = 2$$

Thus  $S(1)$  is true that is condition (I) is satisfied.

(ii) Let us assume that  $S(n)$  is true for any  $n = k \in N$ , i.e.,

$$S(k): 2 + 6 + 18 + \dots + 2 \times 3^{k-1} = 3^k - 1 \quad (A)$$

The given statement for  $n = k + 1$  becomes;

$$2 + 6 + 18 + \dots + 2 \times 3^{k-1} + 2 \times 3^k = 3^{k+1} - 1 \quad (B)$$

Adding  $2 \times 3^k$  on both sides we have

$$2 + 6 + 18 + \dots + 2 \times 3^{k-1} + 2 \times 3^k = 3^k - 1 + 2 \times 3^k$$

$$= 3^k + 2 \times 3^k - 1$$

$$= 3^k (1 + 2) - 1$$

$$= 3 \cdot 3^k - 1$$

$$= 3^{k+1} - 1$$

Hence  $S(k+1)$  is true whenever  $S(k)$  is true. So condition (II) is satisfied.

Therefore both conditions are satisfied, so  $S(n)$  is true  $\forall n \in N$

**Q.8**  $1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + n \times (2n+1) = \frac{n(n+1)(4n+5)}{6}$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + n \times (2n+1) = \frac{n(n+1)(4n+5)}{6} \quad (i)$$

(i) When  $n=1$ , equation (i) becomes;

$$S(1): 1 \times (2 \times 1 + 1) = \frac{1(1+1)(4 \times 1 + 5)}{6}$$

$$S(1): 3 = 3. \text{ thus } S(1) \text{ is true that is condition (I) is satisfied.}$$

(ii) Let us assume that  $S(n)$  is true for any  $n = k \in N$ , i.e.,

$$S(k): 1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + k \times (2k+1) = \frac{k(k+1)(4k+5)}{6} \quad (A)$$

The given statement for  $n = k + 1$  becomes;

$$1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + k \times (2k+1) + (k+1) \times (2k+3) = \frac{(k+1)(k+2)(4k+9)}{6} \quad (B)$$

Adding  $(k+1)(2k+3)$  in (A) we get;

$$1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + k \times (2k+1) + (k+1) \times (2k+3) = \frac{k(k+1)(4k+5)}{6} + (k+1)(2k+3)$$

$$\begin{aligned}
&= (k+1) \left[ \frac{k(4k+5)}{6} + (2k+3) \right] \\
1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + k(2k+1) + (k+1)(2k+3) &= (k+1) \left[ \frac{k(4k+5) + 6(2k+3)}{6} \right] \\
&= (k+1) \left[ \frac{4k^2 + 5k + 12k + 18}{6} \right] \\
&= (k+1) \left[ \frac{4k^2 + 17k + 18}{6} \right] \\
&= (k+1) \left[ \frac{4k^2 + 8k + 9k + 18}{6} \right] \\
&= (k+1) \left[ \frac{4k(k+2) + 9(k+2)}{6} \right] \\
&= \frac{(k+1)(k+2)(4k+9)}{6}
\end{aligned}$$

Which is same as R.H.S of (B)

Hence  $S(k+1)$  is true when  $S(k)$  is true so condition (II) is satisfied.

Therefore both conditions are satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$ .

**Q.9**  $1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n \times (n+1) = \frac{n(n+1)(n+2)}{3}$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n \times (n+1) = \frac{n(n+1)(n+2)}{3} \quad (i)$$

1. When  $n=1$ , equation (i) becomes;

$$S(1): 1 \times (1+1) = \frac{1(1+1)(1+2)}{3}$$

$$S(1): \quad 2 = 2$$

So statement is true for  $n=1$ , that is condition (I) is satisfied.

2. Let us assume that statement is true for  $n=k \in \mathbb{N}$ , i.e.,

$$S(k): 1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3} \quad (A)$$

Give statement for  $n=k+1$  becomes;

$$S(k+1): 1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (k+1) \times (k+2) = \frac{(k+1)(k+2)(k+3)}{3} \quad (B)$$

Adding  $(k+1)(k+2)$  both sides of (A) we get;

$$\begin{aligned} 1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2) &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= (k+1)(k+2) \left( \frac{k}{3} + 1 \right) \\ &= \frac{(k+1)(k+2)(k+3)}{3} \end{aligned}$$

Which is same as R.H.S of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**Q.10**  $1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2n-1) \times 2n = \frac{n(n+1)(4n-1)}{3}$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2n-1)(2n) = \frac{n(n+1)(4n-1)}{3} \quad (i)$$

1. when  $n = 1$ , equation (i) becomes;

$$S(1): (2 \times 1 - 1)(2 \times 1) = \frac{1(1+1)(4 \times 1 - 1)}{3}$$

$$S(1): 2 = 2$$

So statement is true for  $n = 1$  so condition (I) is satisfied.

2. Suppose that statement is true for  $n = k$ , i.e.,

$$S(k): 1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2k-1)(2k) = \frac{k(k+1)(4k-1)}{3} \quad (A)$$

Given statement for  $n = k+1$  becomes;

$$S(k+1): 1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2k-1)(2k) + (2k+1)(2k+2) = \frac{(k+1)(k+2)(4k+3)}{3} \quad (B)$$

Adding  $(2k+1)(2k+2)$  on both sides of (A) we get :

$$\begin{aligned} 1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2k-1)(2k) + (2k+1)(2k+2) \\ &= \frac{k(k+1)(4k-1)}{3} + (2k+1)(2k+2) \\ &= \frac{k(k+1)(4k-1)}{3} + (2k+1) \cdot 2(k+1) \\ &= (k+1) \left[ \frac{k(4k-1)}{3} + 2(2k+1) \right] \\ &= (k+1) \left[ \frac{4k^2 - k + 12k + 6}{3} \right] \end{aligned}$$



$$\begin{aligned} &= (k+1) \left[ \frac{4k^2 + 11k + 6}{3} \right] \\ &= (k+1) \left[ \frac{4k^2 + 8k + 3k + 6}{3} \right] \\ &= (k+1) \left[ \frac{4k(k+2) + 3(k+2)}{3} \right] \\ &= \frac{(k+1)(k+2)(4k+3)}{3} \end{aligned}$$

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**Q.11**  $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1} \quad (i)$$

1. when  $n = 1$ , equation (i) becomes;

$$S(1): \frac{1}{1(1+1)} = 1 - \frac{1}{1+1}$$

$$S(1): \frac{1}{2} = \frac{1}{2}$$

So statement is true for  $n = 1$  so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$

$$S(k): \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} = 1 - \frac{1}{k+1} \quad (A)$$

Given statement for  $n = k+1$  becomes

$$S(k+1): \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{k+2} \quad (B)$$

Adding  $\frac{1}{(k+1)(k+2)}$  on both sides of (A) we get ;

$$\begin{aligned} & \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{(k+1)} + \frac{1}{(k+1)(k+2)} \\ & = 1 - \left[ \frac{(k+2)-1}{(k+1)(k+2)} \right] \\ & = 1 - \left[ \frac{k+1}{(k+1)(k+2)} \right] \\ & = 1 - \frac{1}{k+2} \end{aligned}$$

Which is same as R.H.S of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

$$\text{Q.12} \quad \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1} \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): \frac{1}{(2(1)-1)(2(1)+1)} = \frac{1}{2(1)+1}$$

$$S(1): \frac{1}{3} = \frac{1}{3}$$

Thus statement is true for  $n=1$ , so condition (I) is satisfied.

2. Suppose that statement is true for  $n=k$ , i.e.,

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1} \quad (A)$$

Given statement for  $n=k+1$  becomes

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)} = \frac{(k+1)}{(2k+3)} \quad (B)$$

Adding  $\frac{1}{(2k+1)(2k+3)}$  on both sides of (A) we get ;

$$\begin{aligned} & \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{k}{(2k+1)} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{1}{(2k+1)} \left[ k + \frac{1}{2k+3} \right] \\ &= \frac{1}{(2k+1)} \left[ \frac{2k^2 + 3k + 1}{2k+3} \right] \end{aligned} \quad \begin{aligned} & \frac{2k^2 + 2k + k + 1}{\therefore 2k(k+1) + 1(k+1)} \\ & \frac{(k+1)(2k+1)}{(k+1)(2k+1)} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{(2k+1)} \left[ \frac{2k(k+1) + 1(k+1)}{2k+3} \right] \\ &= \frac{1}{(2k+1)} \left[ \frac{(2k+1)(k+1)}{2k+3} \right] \\ &= \frac{k+1}{2k+3} \end{aligned}$$

Which is same as R.H.S of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

$$\text{Q.13} \quad \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{2(3n+2)}$$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{2(3n+2)} \quad (\text{i})$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): \frac{1}{(2)(5)} = \frac{1}{(2)(5)}$$

So  $S(1)$  is true, so condition (I) is satisfied

2. Suppose that given statement is true for  $n=k$ , i.e.,

$$S(k): \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3k-1)(3k+2)} = \frac{k}{2(3k+2)} \quad (\text{A})$$

Given statement for  $n=k+1$  becomes

$$S(k+1): \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3k-1)(3k+2)} + \frac{1}{(3k+2)(3k+5)} = \frac{k+1}{2(3k+5)} \quad (\text{B})$$

Adding  $\frac{1}{(3k+2)(3k+5)}$  on both sides of (A) we get ;

$$\begin{aligned} & \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3k-1)(3k+2)} + \frac{1}{(3k+2)(3k+5)} \\ &= \frac{k}{2(3k+2)} + \frac{1}{(3k+2)(3k+5)} \\ & \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3k-1)(3k+2)} + \frac{1}{(3k+2)(3k+5)} = \frac{1}{(3k+2)} \left[ \frac{k}{2} + \frac{1}{3k+5} \right] \\ &= \frac{1}{(3k+2)} \left[ \frac{k(3k+5)+2}{2(3k+5)} \right] \\ &= \frac{1}{(3k+2)} \left[ \frac{3k^2+5k+2}{2(3k+5)} \right] \\ &= \frac{1}{(3k+2)} \left[ \frac{3k^2+3k+2k+2}{2(3k+5)} \right] \\ &= \frac{1}{(3k+2)} \left[ \frac{3k(k+1)+2(k+1)}{2(3k+5)} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{(3k+2)} \left[ \frac{(3k+2)(k+1)}{2(3k+5)} \right] \\
&= \frac{(k+1)}{2(3k+5)}
\end{aligned}$$

Which is same as R.H.S of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**Q.14**  $r + r^2 + r^3 + \dots + r^n = \frac{r(1-r^n)}{(1-r)}, (r \neq 1)$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): r + r^2 + r^3 + \dots + r^n = \frac{r(1-r^n)}{(1-r)} \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): r^1 = \frac{r(1-r^1)}{(1-r)}$$

$$S(1): r = r$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$ , i.e.,

$$S(k): r + r^2 + r^3 + \dots + r^k = \frac{r(1-r^k)}{(1-r)} \quad (A)$$

For  $n=k+1$  given statement becomes;

$$S(k+1): r + r^2 + r^3 + \dots + r^k + r^{k+1} = \frac{r(1-r^{k+1})}{(1-r)} \quad (B)$$

Adding  $r^{k+1}$  on both sides of (A) we get ;

$$\begin{aligned}
r + r^2 + r^3 + \dots + r^k + r^{k+1} &= \frac{r(1-r^k)}{(1-r)} + r^{k+1} = r \left( \frac{(1-r^k)}{1-r} + r^k \right) \\
&= r \left( \frac{1-r^k + r^k(1-r)}{(1-r)} \right) \\
&= r \left( \frac{1-r^k + r^k - r^{k+1}}{(1-r)} \right) \\
&= \frac{r(1-r^{k+1})}{(1-r)}
\end{aligned}$$

Which is same as R.H.S of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**Q.15**  $a + (a+d) + (a+2d) + \dots + (a+(n-1)d) = \frac{n}{2} [2a + (n-1)d]$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$\begin{aligned} S(n): & a + (a+b) + (a+2d) + \dots + (a+(n-1)d) \\ &= \frac{n}{2} [2a + (n-1)d] \end{aligned} \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): [a + (1-1)d] = \frac{1}{2} [2a + (1-1)d]$$

$$S(1): a = \frac{1}{2}(2a)$$

$$S(1): a = a$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$ , i.e.,

$$\begin{aligned} S(k): & a + (a+d) + (a+2d) + \dots + (a+(k-1)d) \\ &= \frac{k}{2} [2a + (k-1)d] \quad (A) \end{aligned}$$

For  $n=(k+1)$  given statement becomes;

$$\begin{aligned} S(k+1): & a + (a+d) + (a+2d) + \dots + (a+(k-1)d) + (a+kd) \\ &= \frac{(k+1)}{2} [2a + kd] \quad (B) \end{aligned}$$

Adding  $(a+kd)$  on both sides of (A) we get ;

$$\begin{aligned} & a + (a+d) + (a+2d + \dots +) (a+(k-1)d) + (a+kd) \\ &= \frac{k}{2} [2a + (k-1)d] + (a+kd) \\ & a + (a+d) + (a+2d) + \dots + (a+(k-1)d) + (a+kd) \\ &= ka + \frac{k}{2}(k-1)d + a + kd \\ &= ka + a + \frac{k}{2}(k-1)d + kd \\ &= a(k+1) + kd \left( \frac{(k-1)}{2} + 1 \right) \\ &= a(k+1) + kd \left( \frac{k+1}{2} \right) \\ &= \frac{(k+1)}{2} (2a + kd) \end{aligned}$$

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**Q.16**  $1! + 2! + 3! + \dots + n! = n! + 1 - 1$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1! + 2! + 3! + \dots + n! = n! + 1 - 1 \quad (i)$$

1. when  $n = 1$ , equation (i) becomes;

$$S(1): 1! = 1! + 1 - 1$$

$$S(1): 1 = 2 - 1$$

$$S(1): 1 = 2 - 1 \quad \because 2! = 2$$

$$S(1): 1 = 1$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n = k \in N$ , i.e.,

$$S(k): 1! + 2! + 3! + \dots + k! = k! + 1 - 1 \quad (A)$$

Given statement for  $n = k + 1$  becomes ;

$$S(k+1): 1! + 2! + 3! + \dots + k! + (k+1)! = k! + 1 - 1 \quad (B)$$

Adding  $(k+1) \cdot (k+1)$  on both sides of (A) we get ;

$$\begin{aligned} 1! + 2! + 3! + \dots + k! + (k+1)! &= k! + 1 - 1 + (k+1) \cdot (k+1) \\ &= (k+1) - 1 + (k+1) \cdot (k+1) \\ &= (k+1) [1 + (k+1)] - 1 \\ &= (k+1)(k+2) - 1 \\ &= (k+2) \cdot (k+1) - 1 \\ &= (k+2) - 1 \end{aligned}$$

Which is Same as R.H.S of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true

$\forall n \in N$

**Q.17**  $a_n = a_1 + (n-1)d$  when  $a_1, a_1 + d, a_1 + 2d, \dots$  are in A.P.

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): a_n = a_1 + (n-1)d \quad (i)$$

1. when  $n = 1$ , equation (i) becomes;

$$S(1): a_1 = a_1 + (1-1)d$$

$$S(1): a_1 = a_1$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n = k \in N$ , i.e.,

$$S(k): a_k = a_1 + (k-1)d \quad (A)$$

So given statement for  $n = k+1$  becomes

$$S(k): a_{k+1} = a + kd \quad (B)$$

Adding  $d$  on both sides of (A) we get ;

$$\begin{aligned} a_k + d &= a_1 + (k-1)d + d \\ &= a_1 + kd - d + d \\ &= a_1 + kd = \text{R.H.S of (B)} \end{aligned}$$

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**Q.18**  $a_n = a_1 r^{n-1}$  when  $a_1, a_1 r, a_1 r^2, \dots$  form a G.P.

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): a_n = a_1 r^{n-1} \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): a_1 = a_1 r^{1-1}$$

$$S(1): a_1 = a_1$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$ , i.e.,

$$S(k): a_k = a_1 r^{k-1} \quad (A)$$

So given statement for  $n=k+1$  becomes

$$S(k+1): a_{k+1} = a_1 r^k \quad (B)$$

Multiply  $r$  on both sides of (A) we get ;

$$r.a_k = a_1 r^{k-1} . r$$

$$a_{k+1} = a_1 r^k$$

Which is right hand side of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

$$\text{Q.19 } 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(4n^2-1)}{3}$$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(4n^2-1)}{3} \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): (2 \times 1 - 1)^2 = \frac{1(4 \times 1^2 - 1)}{3}$$

$$S(1): 1 = 1$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$ , i.e.,



$$S(k): 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{k(4k^2-1)}{3} \quad (A)$$

So given statement for  $n = k+1$  becomes

$$\begin{aligned} S(k+1): 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 &= \frac{(k+1)(4(k+1)^2-1)}{3} \\ &= \frac{(k+1)(4k^2+8k+3)}{3} \end{aligned} \quad (B)$$

Adding  $(2k+1)^2$  on both sides of (A) we get;

$$\begin{aligned} 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 &= \frac{k(4k^2-1)}{3} + (2k+1)^2 \\ &= \frac{k(4k^2-1) + 3(2k+1)^2}{3} \\ &= \frac{k((2k-1)(2k+1))}{3} + (2k+1)^2 \\ &= (2k+1) \left[ \frac{k(2k-1)}{3} + (2k+1) \right] \\ &= (2k+1) \left[ \frac{(2k-1) + 3(2k+1)}{3} \right] \\ &= (2k+1) \left[ \frac{2k^2 - k + 6k + 3}{3} \right] \\ &= (2k+1) \left[ \frac{2k^2 + 5k + 3}{3} \right] \\ &= (2k+1) \frac{[2k^2 + 2k + 3k + 3]}{3} \\ &= \frac{(2k+1)(2k+3)(k+1)}{3} \\ &= \frac{(k+1)[4k^2 + 8k + 3]}{3} \end{aligned}$$

Which is right hand side of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true

$\forall n \in \mathbb{N}$

**Q.20**  $\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{n+2}{3} = \binom{n+3}{4}$

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{n+2}{3} = \binom{n+3}{4} \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1): \binom{1+2}{3} = \binom{1+3}{4}$$

$$S(1): \binom{3}{3} = \binom{4}{4}$$

$$S(1): 1=1$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$ , i.e.,

$$S(k): \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{k+2}{3} = \binom{k+3}{4} \quad (A)$$

So given statement for  $n=k+1$  becomes

$$S(k+1): \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{k+2}{3} + \binom{k+3}{3} = \binom{k+4}{4} \quad (B)$$

Adding  $\binom{k+3}{3}$  on both sides of (A) we get ;

$$\begin{aligned} \binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{k+2}{3} + \binom{k+3}{3} &= \binom{k+3}{3} + \binom{k+3}{4} \\ &= \binom{k+4}{4} \quad \because \binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r} \end{aligned}$$

Which is right hand side of (B)

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true

$\forall n \in \mathbb{N}$

**Q.21 Prove by the mathematical induction that for all positive integral value of  $n$ .**

- (i)  $n^2 + n$  is divisible by 2.

**Solution:**

Let  $S(n) = n^2 + n$  be the given statement, i.e.,

$$S(n): n^2 + n \quad (i)$$

1. when  $n=1$ , equation (i) becomes;

$$S(1) = 1^2 + 1 = 2 \text{ that is divisible by 2.}$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$ , i.e.,

$$S(k): k^2 + k \text{ is divisible by 2, that is } \frac{k^2 + k}{2} = Q \text{ where Q is Quotient,}$$

$$\text{i.e., } k^2 + k = 2Q \quad (A)$$

So given statement for  $n=k+1$  becomes

$$\begin{aligned} S(k+1): & (k+1)^2 + (k+1) \\ &= k^2 + 1 + 2k + k + 1 \end{aligned} \quad (B)$$

$$\begin{aligned}
&= (k^2 + k) + (2k + 2) \\
&= 2Q + 2(k + 1) \text{ by using A} \\
&= 2[Q + (k + 1)] \\
&= \text{Which is divisible by 2.}
\end{aligned}$$

Hence  $S(k+1)$  is true if  $S(k)$  is true, so condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in \mathbb{N}$

**(ii)  $5^n - 2^n$  is divisible by 3.**

**Solution:**

Let  $S(n)$  be the given statement, i.e.,

$$S(n): 5^n - 2^n \quad (i)$$

1. when  $n = 1$ , equation (i) becomes;

$$S(1): 5^1 - 2^1$$

$$S(1): 3 \text{ that is divisible by 3}$$

Thus  $S(1)$  is true, so condition (I) is satisfied.

2. Suppose that given statement is true for  $n = k$ , i.e.,

$$S(k): 5^k - 2^k \text{ is divisible by 3, so}$$

$$\frac{5^k - 2^k}{3} = Q \text{ where } Q \text{ is Quotient, i.e.,}$$

$$5^k - 2^k = 3Q \quad (A)$$

Next we have to show that statement is also true for  $n = k + 1$ , that is we have to show that  $S(k + 1) = 5^{k+1} - 2^{k+1}$  is also divisible by 3.

So consider

$$\begin{aligned}
5^{k+1} - 2^{k+1} &= 5^k \cdot 5 - 2^k \cdot 2 \\
&= 5(3Q + 2^k) - 2^{k+1} && \because \text{from (A) } 5^k = 3Q + 2^k \\
&= 15Q + 5 \cdot 2^k - 2^{k+1} \\
&= 15Q + 5 \cdot 2^k - 2^k \cdot 2 \\
&= 15Q + 2^k(5 - 2) \\
&= 15Q + 3 \cdot 2^k \\
&= 3[5Q + 2^k] \text{ that is divisible by } 3 \forall k \in \mathbb{N}.
\end{aligned}$$

Thus statement is true for  $n = k + 1$  when  $S(k)$  is true. So condition (II) is satisfied hence result is true  $\forall n \in \mathbb{N}$ .

**(iii)  $5^n - 1$  is divisible by 4**

**Solution:**

Let the given statement is  $S(n)$  i.e.,

$$S(n): 5^n - 1 \quad (i)$$

1. when  $n = 1$  then equation (i) becomes

$S(1)$ :  $5^1 - 1 = 4$  which is divisible by 4

So the condition (I) is satisfied

2. Let the statement is true for  $n = k$  i.e.,

$S(k)$ :  $5^k - 1^k$  is divisible by 4 i.e.,

$$\frac{5^k - 1}{4} = Q \text{ where } Q \text{ is the Quotient}$$

$$\Rightarrow 5^k - 1 = 4Q \tag{A}$$

Now we have to show that statement is also true for  $n = k + 1$  i.e.,

$S(k+1)$ :  $5^{k+1} - 1$  is also divisible by 4.

So consider

$$\begin{aligned} 5^{k+1} - 1 &= 5 \cdot 5^k - 1 \\ &= 5(4Q + 1) - 1 \\ &= 20Q + 5 - 1 && \because \text{from (A)} \\ &= 20Q + 4 && 5^k = 4Q + 1 \\ &= 4(5Q + 1) \text{ which is divisible by 4} \end{aligned}$$

Thus  $S(k+1)$  is true whenever  $S(k)$  is true.

Hence result is true  $\forall n \in N$ .

(iv)  $8 \times 10^n - 2$  is divisible by 6.

**Solution:**

Let the given statement is  $S(n)$  i.e.,

$$S(n): 8 \times 10^n - 2 \quad (i)$$

1. For  $n=1$ , equation (i) becomes  
 $S(1): 8 \times 10^1 - 2 = 78$  that is divisible by 6.
2. Suppose that given statement is true for  $n=k$ , i.e.,  
 $S(k): 8 \times 10^k - 2$  is divisible by 6, i.e.,

$$\frac{8 \times 10^k - 2}{6} = Q \text{ where } Q \text{ is Quotient}$$

$$8 \times 10^k - 2 = 6Q \quad (A)$$

Now we have to show that statement is also true for  $n=k+1$ , i.e.,

$$S(k+1): 8 \times 10^{k+1} - 2 \text{ is also divisible by 6.}$$

So consider

$$\begin{aligned} 8 \times 10^{k+1} - 2 &= 8 \times 10^k \times 10 - 2 \\ &= 10(8 \times 10^k - 2) - 2 \\ &\quad \because \text{from (A)} \\ &= 10(6Q) - 2 \quad \quad \quad 8 \times 10^k - 2 = 6Q \\ &= 60Q - 2 \\ &= 6(10Q - \frac{2}{6}) \\ &= 6(10Q - \frac{1}{3}) \end{aligned}$$

Thus  $S(k+1)$  is true whenever  $S(k)$  is true. Hence  $S(n)$  is true  $\forall n \in N$ .

(v)  $n^3 - n$  is divisible by 6

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): n^3 - n \quad (i)$$

1. When  $n=1$  then equation (i) will become ;  
 $S(1): 1^3 - 1 = 0$  which is divisible by 6.
2. Suppose that given statement is true for  $n=k$  i.e.,  
 $S(k): k^3 - k$  is divisible by 6, so

$$\frac{k^3 - k}{6} = Q \text{ where } Q \text{ is quotient}$$

$$k^3 - k = 6Q \quad (A)$$

Now we have to show that statement is also true for  $n=k+1$  i.e.,

$$S(k+1): (k+1)^3 - (k+1)$$

So consider

$$(k+1)^3 - (k+1) = k^3 + 1 + 3k^2 + 3k - k - 1$$

$$\begin{aligned}
&= (k^3 - k) + 3k^2 + 3k \\
&= 6Q + 3(k^2 + k) && \because k^3 - k = 6Q \\
& && \therefore x^2 + x \text{ is divisible by 2} \\
&= 6Q + 3k(k+1) \\
&= 6Q + 3(2p) && \because k \in N. \text{ so} \\
&= 6Q + P && k(k+1) = \text{an even integer} \\
&= 6(Q + P) \text{ which is divisible by 6}
\end{aligned}$$

Thus  $S(k+1)$  is true whenever  $S(k)$  is true so given statement is true  $\forall n \in N$ .

**Q.22**  $\frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n} = \frac{1}{2} \left[ 1 - \frac{1}{3^n} \right]$

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n} = \frac{1}{2} \left[ 1 - \frac{1}{3^n} \right] \quad (i)$$

1. when  $n=1$ , equation (i) becomes ;

$$\begin{aligned}
S(1): \frac{1}{3^1} &= \frac{1}{2} \left[ 1 - \frac{1}{3^1} \right] \\
\frac{1}{3} &= \frac{1}{2} \left[ \frac{2}{3} \right] \\
\frac{1}{3} &= \frac{1}{3}
\end{aligned}$$

So  $S(1)$  is true so condition (I) is satisfied.

2. Suppose that given statement is true for  $n=k$  i.e.,

$$S(k): \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^k} = \frac{1}{2} \left[ 1 - \frac{1}{3^k} \right] \quad (A)$$

Given statement for  $n=k+1$  becomes ;

$$S(k+1): \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^k} + \frac{1}{3^{k+1}} = \frac{1}{2} \left[ 1 - \frac{1}{3^{k+1}} \right] \quad (B)$$

Adding  $\frac{1}{3^{k+1}}$  on both side of (A) , we get ;

$$\begin{aligned}
\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^k} + \frac{1}{3^{k+1}} &= \frac{1}{2} \left[ 1 - \frac{1}{3^k} \right] + \frac{1}{3^{k+1}} \\
&= \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{3^k} + \frac{1}{3} \cdot \frac{1}{3^k} \\
&= \frac{1}{2} - \left( \frac{1}{2} - \frac{1}{3} \right) \cdot \frac{1}{3^k} \\
&= \frac{1}{2} - \frac{1}{6} \cdot \frac{1}{3^k}
\end{aligned}$$

$$= \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{3^k}$$

$$= \frac{1}{2} \left[ 1 - \frac{1}{3^{k+1}} \right]$$

Which is right hand side of (B)

Thus  $S(k+1)$  is true whenever  $S(k)$  is true. So condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in N$ .

**Q.23**  $1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n-1} \cdot n^2 = \frac{(-1)^{n-1} \cdot n(n+1)}{2}$

**Solution:**

Let the given statement is  $S(n)$ , i.e.,

$$S(n): 1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n-1} \cdot n^2 = \frac{(-1)^{n-1} \cdot n(n+1)}{2} \quad (i)$$

1. when  $n=1$  then equation (i) becomes ;

$$S(1): (-1)^{1-1} \cdot (1)^2 = \frac{(-1)^{1-1} \cdot (1)(1+1)}{2}$$

$$S(1): 1 = 2$$

$$S(1): 1 = 1$$

So  $S(1)$  is true and condition (I) is satisfied

2. Suppose that given statement is true for  $n = k$  i.e.,

$$S(k): 1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{k-1} \cdot (k)^2 = \frac{(-1)^{k-1} \cdot k(k+1)}{2} \quad (A)$$

Given statement for  $n = k+1$  becomes

$$S(k+1): 1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^k (k+1)^2 = \frac{(-1)^k \cdot (k+1)(k+2)}{2} \quad (B)$$

By adding

$(-1)^k (k+1)^2$  on both sides of (A) we get ;

$$1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{k-1} k^2 + (-1)^k (k+1)^2 = \frac{(-1)^{k-1} \cdot k(k+1)}{2} + (-1)^k (k+1)^2$$

$$= (-1)^k (k+1) \left[ \frac{(-1)^{-1} \cdot k}{2} + (k+1) \right]$$

$$= (-1)^k (k+1) \left[ \frac{-k}{2} + k+1 \right]$$

$$= (-1)^k (k+1) \left[ \frac{-k+2k+2}{2} \right]$$

$$= (-1)^k (k+1) \left[ \frac{k+2}{2} \right]$$



$$= \frac{(-1)^k (k+1)(k+2)}{2}$$

Which is right hand side of (B)

Thus  $S(k+1)$  is true whenever  $S(k)$  is true. So condition (II) is satisfied, so  $S(n)$  is true  $\forall n \in N$ .

**Q.24**  $1^3 + 3^3 + 5^3 + \dots + (2n-1)^3 = n^2(2n^2-1)$

**Solution:**

Let the given statement is  $S(n)$ , i.e.,

$$S(n) = 1^3 + 3^3 + 5^3 + \dots + (2n-1)^3 = n^2(2n^2-1) \quad (i)$$

1. When  $n = 1$  then equation (i) becomes

$$S(1): (2(1)-1)^3 = 1^2(2(1)^2-1)$$

$$S(1): 1 = 1$$

Thus  $S(1)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): 1^3 + 3^3 + 5^3 + \dots + (2k-1)^3 = k^2(2k^2-1) \quad (A)$$

Given statement for  $n = k+1$  becomes ;

$$\begin{aligned} S(k+1): 1^3 + 3^3 + 5^3 + \dots + (2k-1)^3 + (2k+1)^3 &= (k+1)^2(2(k+1)^2-1) \\ &= (k+1)^2(2k^2+4k+1) \end{aligned} \quad (B)$$

Adding  $(2k+1)^3$  on both sides of (A) we get;

$$\begin{aligned} 1^3 + 3^3 + 5^3 + \dots + (2k-1)^3 + (2k+1)^3 &= k^2(2k^2-1) + (2k+1)^3 \\ &= 2k^4 - k^2 + (8k^3 + 1 + 12k^2 + 6k) \\ &= 2k^4 + 8k^3 + 11k^2 + 6k + 1 \\ &= 2k^4 + 2k^3 + 6k^3 + 6k^2 + 5k^2 + 5k + k + 1 \\ &= 2k^3(k+1) + 6k^2(k+1) + 5k(k+1) + (k+1) \\ &= (k+1)[2k^3 + 6k^2 + 5k + 1] \\ &= (k+1)[2k^3 + 2k^2 + 4k^2 + 4k + k + 1] \\ &= (k+1)[2k^2(k+1) + 4k(k+1) + (k+1)] \\ &= (k+1)^2[2k^2 + 4k + 1] \end{aligned}$$

Which is right hand side of (B)

Thus  $S(k+1)$  is true whenever  $S(k)$  is true. So condition (II) is satisfied

Hence  $S(n)$  is true  $\forall n \in N$ .

**Q.25**  $(x+1)$  is factor of  $x^{2n}-1; (x \neq -1)$

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): x^{2n} - 1 \quad (i)$$

1. When  $n = 1$ , then equation (i) becomes

$$S(1): x^{2(1)} - 1 = x^2 - 1 \text{ which is divisible by } x + 1$$

So (1) is true, so condition (I) is satisfied.

2. Suppose that statement is true for  $n = k$  that is  $S(k)$  is divisible by  $(x + 1)$ . So

$$\frac{x^{2k} - 1}{x + 1} = Q \text{ where } Q \text{ is Quotient}$$

$$x^{2k} - 1 = Q(x + 1) \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$S(k + 1)$  is also divisible by  $(x + 1)$ .

consider  $S(k + 1): x^{2(k+1)} - 1$

So

$$\begin{aligned} x^{2k+2} - 1 &= x^{2k} \cdot x^2 - 1 \\ &= x^2 [Q(x + 1) + 1] - 1 \quad \because \text{from (A)} \quad x^{2k} = Q(x + 1) + 1 \\ &= x^2 \cdot Q(x + 1) + (x^2 - 1) \\ &= x^2 \cdot Q(x + 1) + (x - 1)(x + 1) \\ &= (x + 1) [x^2 \cdot Q + (x - 1)] \end{aligned}$$

Which is divisible by  $(x + 1)$ .

Thus  $S(k + 1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied. Hence  $S(n)$  is true  $\forall n \in N$ .

### Q.26 $(x - y)$ is a factor of $x^n - y^n$ ; $(x \neq y)$

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): x^n - y^n \quad (i)$$

1. When  $n = 1$  then equation (i) becomes

$$S(1): x^1 - y^1 \text{ which is divisible by } (x - y).$$

So  $S(1)$  is true and condition (I) is satisfied.

2. Suppose that statement is true  $n = k$  i.e.,

$$S(k): x^k - y^k \text{ is divisible by } x - y$$

i.e.,

$$\frac{x^k - y^k}{x - y} = Q \text{ where } Q \text{ is Quotient}$$

$$x^k - y^k = Q(x - y) \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$S(k+1)$  is also divisible by  $(x+1)$

So consider

$$\begin{aligned}
 S(k+1): x^{k+1} - y^{k+1} &= x^k \cdot x - y^{k+1} \\
 &= x[Q(x-y) + y^k] - y^{k+1} \quad \because \text{from (A)} \quad x^k = Q(x-y) + y^k \\
 &= x.Q(x-y) + x.y^k - y^k.y \\
 &= x.Q(x-y) + y^k(x-y) \\
 &= (x-y)[x.Q + y^k]
 \end{aligned}$$

Which is divisible by  $(x-y)$ . thus  $S(k+1)$  is true. Whenever  $S(k)$  is true. So condition (II) is satisfied, so the given statement is true  $\forall n \in N$ .

**Q.27**  $(x+y)$  is a factor of  $x^{2n-1} + y^{2n-1}$  ( $x \neq -y$ )

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$S(n): (x+y)$  is a factor of  $x^{2n-1} + y^{2n-1}$

1. When  $n=1$  then  $S(n)$  becomes

$S(1): x^{2(1)-1} + y^{2(1)-1} = x + y$  so which is divisible by  $x + y$  so  $S(1)$  is true and condition (I) is satisfied.

2. Suppose that statement is true for  $n = k$  that is

$$\frac{x^{2k-1} + y^{2k-1}}{x+y} = Q$$

$$x^{2k-1} + y^{2k-1} = Q(x+y) \quad (A)$$

Now we show that statement is also true for  $n = k+1$  i.e.,

So consider

$$\begin{aligned}
 x^{2k+1} + y^{2k+1} &= x^{2k-1} \cdot x^2 + y^{2k-1} \cdot y^2 \\
 &= x^2[Q(x+y) - y^{2k-1}] + y^{2k-1} \cdot y^2 \quad \because \text{from (A)} \quad x^{2k-1} = Q(x+y) - y^{2k-1} \\
 &= x^2.Q(x+y) - x^2.y^{2k-1} + y^{2k-1}y^2 \\
 &= x^2Q(x+y) - y^{2k-1}[x^2 - y^2] \\
 &= x^2.Q(x+y) - y^{2k-1}(x-y)(x+y) \\
 &= (x+y)[x^2.Q - (x-y).y^{2k-1}]
 \end{aligned}$$

Which is divisible by  $(x+y)$  thus  $S(k+1)$  is true whenever  $S(k)$  is true so condition (II) is satisfied.

Hence  $S(n)$  is true  $\forall n \in N$ .

**Q.28** Use mathematical induction to show that

$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$  for all non-negative integers  $n$ .

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): 1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1 \quad (i)$$

1. When  $n = 0$  then equation (i) becomes

$$S(0): 2^0 = 2^{0+1} - 1 \quad \text{Here } \because n \in W$$

$$S(0): 1 = 2^1 - 1$$

$$S(0): 1 = 1$$

So  $S(0)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): 1 + 2 + 2^2 + 2^3 + \dots + 2^k = 2^{k+1} - 1 \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): 1 + 2 + 2^2 + 2^3 + \dots + 2^k + 2^{k+1} = 2^{k+2} - 1 \quad (B)$$

In order to prove (B) we add  $2^{k+1}$  on both sides of (A) we get

$$\begin{aligned} 1 + 2 + 2^2 + 2^3 + \dots + 2^k + 2^{k+1} &= 2^{k+1} - 1 + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1 \end{aligned}$$

Which is right hand side of (B)

Thus  $S(k+1)$  is true whenever  $S(k)$  is true so condition (II) is satisfied.

Hence  $S(n)$  is true  $\forall n \in N$ .

**Q.29 If A and B are square matrices and  $AB = BA$ , then show by mathematical Induction that  $AB^n = B^n A$  for any positive integer n.**

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): AB^n = B^n A \quad (i)$$

1. When  $n = 1$  then  $S(n)$  becomes

$$S(1): AB^1 = B^1 A$$

$$S(1): AB = BA$$

So  $S(1)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): AB^k = B^k A \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): AB^{k+1} = B^{k+1} A \quad (B)$$

Multiply matrix B from left side with (A) we get

$$B(AB^k) = B.B^k A$$

$$(BA)B^k = B^{k+1} A \quad \text{By associative law}$$

$$(AB)B^k = B^{k+1} A \quad AB = BA \text{ (Given)}$$

$$A(BB^k) = B^{k+1} A \quad \text{By associative law}$$

$$AB^{k+1} = B^{k+1} A$$

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n \in N$ .

**Q.30 Prove by the principle of mathematical induction that  $n^2 - 1$  is divisible by 8, when  $n$  an odd integer.**

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): n^2 - 1 \text{ is divisible by } 8 \quad (i)$$

1. When  $n=1$  then equation (i) becomes

$$S(1) = 1^2 - 1$$

$$S(1) = 0 \text{ which is divisible by } 8$$

So  $S(1)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): \frac{k^2 - 1}{8} = Q \Rightarrow k^2 - 1 = 8Q \quad (A)$$

Where Q is quotient.

Now we show that statement is also true for  $n = k+2$  i.e.,

$$S(k+2): (k+2)^2 - 1 \quad (B)$$

So consider,

$$\begin{aligned} (k+2)^2 - 1 &= k^2 + 4k + 4 - 1 \\ &= (k^2 - 1) + (4k + 4) \\ &= 8Q + 4(k+1) \\ &= 8Q + 4(2p) \quad \therefore k \in O, p \in N \\ &= 8[Q + p] \end{aligned}$$

Which is divisible by 8

So  $S(k+2)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n \in N$ .

**Q.31 Use the principle of mathematical induction to prove that  $\ln x^n = n \ln x$  for any positive integer  $n \geq 0$  if  $x$  is positive integer.**

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): \ln x^n = n \ln x \quad (i)$$

1. When  $n=0$  then equation (i) becomes

$$S(0): \ln x^0 = 0 \cdot \ln x$$

$$S(0): \ln 1 = 0$$

$$S(0): 0 = 0$$

So  $S(0)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): \ln x^k = k \cdot \ln x \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): \ln x^{k+1} = (k+1) \cdot \ln x \quad (B)$$

So in order to prove (B) adding  $\ln x$  on both sides of (A) we get;

$$\ln x^k + \ln x = k \cdot \ln x + \ln x$$

$$\ln(x^k \cdot x) = (k+1) \ln x$$

$$\ln x^{k+1} = (k+1) \ln x$$

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n \in N$ .

Use the Principle of extended mathematical Induction to prove that

**Q.32**  $n! > 2^n - 1$  for integral values of  $n \geq 4$

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): n! > 2^n - 1 \quad (i)$$

1. When  $n = 4$  then  $S(n)$  becomes

$$S(4): 4! > 2^4 - 1$$

$$S(4): 24 > 15$$

So  $S(4)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): k! > 2^k - 1 \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): (k+1)! > 2^{k+1} - 1 \quad (B)$$

Multiply  $(k+1)$  on both sides of (A) we get ;

$$(k+1).k! > (k+1)(2^k - 1)$$

$$(k+1)! > 2(2^k - 1) \quad \therefore k > 4$$

$$(k+1)! > 2^{k+1} - 2 \quad \therefore k+1 > 2$$

$$(k+1)! > (2^{k+1} - 1) - 1$$

$$(k+1)! > 2^{k+1} - 1$$

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true

$$\forall n \geq 4, n \in N.$$

**Q.33**  $n^2 > n + 3$  for integral values of  $n \geq 3$

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): n^2 > n + 3 \quad (i)$$

1. When  $n = 3$  then  $S(n)$  becomes

$$S(3): 3^2 > 3 + 3$$

$$S(3): 9 > 6$$

So  $S(3)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): k^2 > k + 3 \text{ where } k \geq 3 \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): (k+1)^2 > k + 4 \quad (B)$$

Adding  $2k + 1$  in (A) on both sides

$$k^2 + 2k + 1 > k + 3 + 2k + 1$$

$$(k+1)^2 > (k+4) + 2k$$

$$(k+1)^2 > (k+4) \quad \therefore k \geq 3$$

So  $2k$  is positive integer, so by neglecting  $2k$  L.H.S. become more large.

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true

$\forall n \geq 3, n \in N$ .

**Q.34**  $4^n > 3^n + 2^{n-1}$  for integral values of  $n \geq 2$ .

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): 4^n > 3^n + 2^{n-1} \quad (i)$$

1. When  $n = 2$  then  $S(n)$  becomes

$$S(2): 4^2 > 3^2 + 2^{2-1}$$

$$S(2): 16 > 9 + 2$$

$$S(2): 16 > 11$$

So  $S(2)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): 4^k > 3^k + 2^{k-1} \quad (A)$$

Now we show that statement is also true for  $n = k+1$  i.e.,

$$S(k+1): 4^{k+1} > 3^{k+1} + 2^{k+1-1}$$

$$S(k+1): 4^{k+1} > 3^{k+1} + 2^k \quad (B)$$

In order to prove (2) we multiply (A) by 4 on both sides we get ;

$$4 \cdot 4^k > 4(3^k + 2^{k-1})$$

$$4^{k+1} > 4 \cdot 3^k + 4 \cdot 2^{k-1}$$

$$4^{k+1} > (3+1) \cdot 3^k + (2+2) 2^{k-1}$$

$$4^{k+1} > (3 \cdot 3^k + 2^k) + (3^k + 2^k)$$

$$4^{k+1} > 3^{k+1} + 2^k$$

$3^k + 2^k$  is always positive so by neglecting it, L.H.S become more larger.

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true

$\forall n \geq 2, n \in N$ .

**Q.35**  $3^n < n!$  for integral values of  $n > 6$ .

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): 3^n < n! \quad (i)$$

1. When  $n > 6$ , suppose then for  $n = 7$  then  $S(n)$  becomes

$$S(7): 3^7 < 7!$$

$$S(7): 2187 < 5040$$



So  $S(7)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): 3^k < k! \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): 3^{k+1} < (k+1)! \quad \forall k > 6 \quad (B)$$

In order to prove (1) we multiply  $(k+1)$  on both sides of (A) we get ;

$$(k+1)3^k < (k+1)k! \quad \because k+1 > 0$$

$$3 \cdot 3^k < (k+1)3^k < (k+1)! \quad \because k+1 > 3$$

$$\Rightarrow 3^{k+1} < (k+1)!$$

Which is required as in the equation B

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n > 6$ . where  $n \in N$ .

**Q.36  $n! > n^2$  for integral value of  $n \geq 4$ .**

**Solution:**

Let  $S(n)$  be the given statement i.e.,

$$S(n): n! > n^2 \quad (i)$$

1. When  $n = 4$  then  $S(n)$  becomes

$$S(4): 4! > 4^2 \Rightarrow 24 > 16$$

So  $S(4)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): k! > k^2 \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k+1): (k+1)! > (k+1)^2 \quad (B)$$

In order to prove (2) we multiple  $(k+1)$  on both sides of (A), we get;

$$(k+1)k! > (k+1)k^2 \quad \therefore k^2 > k+1 \quad \forall k \geq 4$$

$$(k+1)! > (k+1)(k+1)$$

$$(k+1)! > (k+1)^2$$

So  $S(k+1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n \geq 4$ . where  $n \in N$ .

**Q.37  $3+5+7+\dots+(2n+5) = (n+2)(n+4)$  for integral values of  $n \geq -1$ .**

**Solution:** Let  $S(n)$  be the given statement i.e.,

$$S(n): 3+5+7+\dots+(2n+5) = (n+2)(n+4) \quad (i)$$

1. When  $n = -1$  then equation (i) becomes

$$S(-1): 2(-1)+5 = (-1+2)(-1+4)$$

$$S(-1): 3 = (1)(3)$$

$$S(-1): 3 = 3$$

So  $S(-1)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): 3 + 5 + 7 + \dots + (2k + 5) = (k + 2)(k + 4) \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k + 1): 3 + 5 + 7 + \dots + (2k + 5) + (2k + 7) = (k + 3)(k + 5) \quad (B)$$

In order to prove (B) we add  $(2k + 7)$  on both sides of (A) we get ;

$$\begin{aligned} 3 + 5 + 7 + \dots + (2k + 5) + (2k + 7) &= (k + 2)(k + 4) + (2k + 7) \\ &= k^2 + 6k + 8 + 2k + 7 \\ &= k^2 + 8k + 15 \\ &= k^2 + 5k + 3k + 15 \\ &= k(k + 5) + 3(k + 5) \\ &= (k + 3)(k + 5) \end{aligned}$$

So  $S(k + 1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n \geq -1$ , where  $n \in \mathbb{Z}$ .

**Q.38**  $1 + nx \leq (1 + x)^n$  for  $n \geq 2$  and  $x > -1$ .

**Solution:** Let  $S(n)$  be the given statement i.e.,

$$S(n): 1 + nx \leq (1 + x)^n \quad (i)$$

1. When  $n \geq 2$  then for  $n = 2$ ,  $S(n)$  becomes

$$S(2): 1 + 2x \leq (1 + x)^2$$

$$S(2): 1 + 2x \leq 1 + x^2 + 2x$$

So  $S(2)$  is true, so condition (I) is satisfied

2. Suppose that statement is true for  $n = k$  i.e.,

$$S(k): 1 + kx \leq (1 + x)^2 \quad (A)$$

Now we show that statement is also true for  $n = k + 1$  i.e.,

$$S(k + 1): 1 + (k + 1)x \leq (1 + x)^{k+1} \quad (B)$$

In order to prove (B) we multiply  $(x + 1)$  on both sides of (A) we get ;

$$(1 + kx)(1 + x) \leq (1 + x)(1 + x)^k$$

$$1 + x + kx + kx^2 \leq (1 + x)^{k+1}$$

$$1 + (k + 1)x + kx^2 \leq (1 + x)^{k+1}$$

$$1 + (k + 1)x \leq (1 + x)^{k+1}$$

So  $S(k + 1)$  is true whenever  $S(k)$  is true, so condition (II) is satisfied, hence  $S(n)$  is true  $\forall n \geq 2$ , where  $n \in \mathbb{N}$ .

**Binomial Theorem:**

An algebraic expression consisting of two terms such as  $a + x$ ,  $x - 2y$ ,  $ax + b$  etc. is called a binomial or a binomial expression e.g.

$$(a + x)^2 = a^2 + 2ax + x^2 \quad (i)$$

$$(a + x)^3 = a^3 + 3a^2x + 3ax^2 + x^3 \quad (ii)$$

The right side of (i) and (ii) are called binomial expansions of binomial  $a + x$  for the indices 2 and 3 respectively.

In general,

$$(a + x)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}x + \binom{n}{2}a^{n-2}x^2 + \dots + \binom{n}{r-1}a^{n-(r-1)}x^{r-1} + \binom{n}{r}a^{n-r}x^r + \dots + \binom{n}{n-1}ax^{n-1} + \binom{n}{n}x^n$$

Or

$$(a + x)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} x^r$$

Where  $a$  and  $x$  are real numbers

In the expansion of  $(a + x)^n$  following points can be observed.

1. The number of terms in the expansion is one greater than its index.
2. The sum of exponents of  $a$  and  $x$  in each term of the expansion is equal to its index.
3. The exponent of  $a$  decreases from index to zero.
4. The exponent of  $x$  increases from zero to index.
5. The coefficients of the terms equidistant from beginning and end of the expansion are

equal as  $\binom{n}{r} = \binom{n}{n-r}$

6. The  $(r+1)^{th}$  term in the expansion is  $\binom{n}{r} a^{n-r} x^r$  and we denote it as  $T_{r+1}$

i.e.,

$$T_{r+1} = \binom{n}{r} a^{n-r} x^r$$

**Middle term in the expansion of  $(a + x)^n$** 

In the expansion of  $(a + x)^n$ , the total number of terms are  $n + 1$ .

**Case-I**

( $n$  is even)

If  $n$  is even then  $n + 1$  is odd,

So  $\left(\frac{n}{2} + 1\right)^{th}$  term will be the only middle term in the expansion.

**Case-II**

( $n$  is odd)

If  $n$  is odd then  $n + 1$  is even,

So  $\left(\frac{n+1}{2}\right)^{th}$  and  $\left(\frac{n+3}{2}\right)^{th}$  terms of the expansion will be the two middle terms.

**Note:**

The sum of coefficients in the expansion of  $(1+x)^n$  is  $2^n$ .

The sum of odd coefficients of binomial expansion = The sum of its even coefficients of binomial expansion =  $2^{n-1}$ .

**EXERCISE 8.2****Q.1 Using binomial theorem to expand the following**

**(i)**  $(a + 2b)^5$

**Solution:**

$$\begin{aligned}
&= {}^5C_0(a)^5(2b)^0 + {}^5C_1(a)^4(2b)^1 + {}^5C_2(a)^3(2b)^2 + {}^5C_3(a)^2(2b)^3 + {}^5C_4(a)^1(2b)^4 + {}^5C_5(a)^0(2b)^5 \\
&= a^5 + 5.a^4.2b + \frac{20}{2}.a^3.4b^2 + 10.a^2.8b^3 + 5.a^1.16b^4 + 32b^5 \\
&= a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5
\end{aligned}$$

**(ii)**  $\left(\frac{x}{2} - \frac{2}{x^2}\right)^6$

**Solution:**

$$\begin{aligned}
&= {}^6C_0\left(\frac{x}{2}\right)^6\left(-\frac{2}{x^2}\right)^0 + {}^6C_1\left(\frac{x}{2}\right)^5\left(-\frac{2}{x^2}\right)^1 + {}^6C_2\left(\frac{x}{2}\right)^4\left(-\frac{2}{x^2}\right)^2 + {}^6C_3\left(\frac{x}{2}\right)^3\left(-\frac{2}{x^2}\right)^3 \\
&+ {}^6C_4\left(\frac{x}{2}\right)^2\left(-\frac{2}{x^2}\right)^4 + {}^6C_5\left(\frac{x}{2}\right)^1\left(-\frac{2}{x^2}\right)^5 + {}^6C_6\left(\frac{x}{2}\right)^0\left(-\frac{2}{x^2}\right)^6 \\
&= \frac{x^6}{64} + 6\left(\frac{x^5}{32}\right)\left(-\frac{2}{x^2}\right) + 15\left(\frac{x^4}{16}\right)\left(\frac{4}{x^4}\right) + 6\left(\frac{x}{2}\right)\left(\frac{-32}{x^{10}}\right) + \left(\frac{64}{x^2}\right) + 20\left(\frac{x^3}{8}\right)\left(-\frac{8}{x^6}\right) + 15\left(\frac{x^2}{4}\right)\left(\frac{16}{x^8}\right) \\
&= \frac{x^6}{64} - \frac{3}{8}x^3 + \frac{15}{4} - \frac{20}{x^3} + \frac{60}{x^6} - \frac{96}{x^9} + \frac{64}{x^{12}}
\end{aligned}$$

**(iii)**  $\left(3a - \frac{x}{3a}\right)^4$

**Solution:**

$$\begin{aligned}
&= {}^4C_0(3a)^4\left(\frac{-x}{3a}\right)^0 + {}^4C_1(3a)^3\left(\frac{-x}{3a}\right)^1 + {}^4C_2(3a)^2\left(\frac{-x}{3a}\right)^2 + {}^4C_3(3a)^1\left(\frac{-x}{3a}\right)^3 + {}^4C_4(3a)^0\left(\frac{-x}{3a}\right)^4 \\
&= (81a^4) + 4(27a^3)\left(\frac{-x}{3a}\right) + 6(9a^2)\left(\frac{x^2}{9a^2}\right) + 4(3a)\left(\frac{-x^3}{27a^3}\right) + \frac{x^4}{81a^4} \\
&= 81a^4 - 36a^2x + 6x^2 - \frac{4x^3}{9a^2} + \frac{x^4}{81a^4}
\end{aligned}$$

**(iv)**  $\left(\frac{x}{2y} - \frac{2y}{x}\right)^8$

**Solution:**

$$\begin{aligned}
&= {}^8C_0\left(\frac{x}{2y}\right)^8\left(\frac{-2y}{x}\right)^0 + {}^8C_1\left(\frac{x}{2y}\right)^7\left(\frac{-2y}{x}\right)^1 + {}^8C_2\left(\frac{x}{2y}\right)^6\left(\frac{-2y}{x}\right)^2 + {}^8C_3\left(\frac{x}{2y}\right)^5\left(\frac{-2y}{x}\right)^3 \\
&+ {}^8C_4\left(\frac{x}{2y}\right)^4\left(\frac{-2y}{x}\right)^4 + {}^8C_5\left(\frac{x}{2y}\right)^3\left(\frac{-2y}{x}\right)^5 + {}^8C_6\left(\frac{x}{2y}\right)^2\left(\frac{-2y}{x}\right)^6 + {}^8C_7\left(\frac{x}{2y}\right)^1\left(\frac{-2y}{x}\right)^7 + {}^8C_8\left(\frac{x}{2y}\right)^0\left(\frac{-2y}{x}\right)^8
\end{aligned}$$

$$\begin{aligned}
&= \frac{x^8}{256y^8} + 8 \left( \frac{x^7}{128y^7} \right) \left( \frac{-2y}{x} \right) + 28 \left( \frac{x^6}{64y^6} \right) \left( \frac{4y^2}{x^2} \right) + 56 \left( \frac{x^5}{32y^5} \right) \left( \frac{-8y^3}{x^3} \right) + 28 \left( \frac{x^2}{4y^2} \right) \left( \frac{64y^6}{x^6} \right) \\
&+ 8 \left( \frac{x}{2y} \right) \left( \frac{-128y^7}{x^7} \right) + \left( \frac{256y^8}{x^8} \right) \\
&= \frac{x^8}{256y^8} - \frac{x^6}{8y^6} + \frac{7x^4}{4y^4} - \frac{14x^2}{y^2} + 70 - 224 \frac{y^2}{x^2} + 448 \frac{y^4}{x^4} - 512 \frac{y^6}{x^6} + \frac{256y^8}{x^8} \\
&= \frac{x^8}{256y^8} - \frac{x^6}{8y^6} + \frac{7x^4}{4y^4} - \frac{14x^2}{y^2} + 70 - 224 \frac{y^2}{x^2} + 448 \frac{y^4}{x^4} - 512 \frac{y^6}{x^6} + \frac{256y^8}{x^8}
\end{aligned}$$

(v)  $\left( \sqrt{\frac{a}{x}} - \sqrt{\frac{x}{a}} \right)^6$

**Solution:**

$$\begin{aligned}
&= {}^6C_0 \left( \sqrt{\frac{a}{x}} \right)^6 \left( -\sqrt{\frac{x}{a}} \right)^0 + {}^6C_1 \left( \sqrt{\frac{a}{x}} \right)^5 \left( -\sqrt{\frac{x}{a}} \right)^1 + {}^6C_2 \left( \sqrt{\frac{a}{x}} \right)^4 \left( -\sqrt{\frac{x}{a}} \right)^2 + {}^6C_3 \left( \sqrt{\frac{a}{x}} \right)^3 \left( -\sqrt{\frac{x}{a}} \right)^3 \\
&+ {}^6C_4 \left( \sqrt{\frac{a}{x}} \right)^2 \left( -\sqrt{\frac{x}{a}} \right)^4 + {}^6C_5 \left( \sqrt{\frac{a}{x}} \right)^1 \left( -\sqrt{\frac{x}{a}} \right)^5 + {}^6C_6 \left( \sqrt{\frac{a}{x}} \right)^0 \left( -\sqrt{\frac{x}{a}} \right)^6 \\
&= \frac{a^3}{x^3} + 6 \left( \frac{a^{\frac{5}{2}}}{x^{\frac{5}{2}}} \right) - \left( \frac{x^2}{a^2} \right) + 15 \left( \frac{a}{x} \right)^2 \left( \frac{x}{a} \right)^1 + 20 \left( \frac{a}{x} \right)^{\frac{3}{2}} \left( \frac{-x}{a} \right)^{\frac{3}{2}} + 15 \left( \frac{a}{x} \right) \left( \frac{x}{a} \right)^2 + 6 \left( \frac{a}{x} \right)^{\frac{1}{2}} \left( \frac{-x}{a} \right)^{\frac{5}{2}} + \left( \frac{x}{a} \right)^3 \\
&= \frac{a^3}{x^3} - \frac{6a^2}{x^2} + 15 \frac{a}{x} - 20 + 15 \frac{x}{a} - 6 \frac{x^2}{a^2} + \frac{x^3}{a^3}
\end{aligned}$$

**Q.2** Calculate the following by means of binomial theorem:

(i)  $(0.97)^3$

**Solution:**

$$\begin{aligned}
&(0.97)^3 \\
&= (1 - 0.03)^3 \\
&= {}^3C_0 (1)^3 (-0.03)^0 + {}^3C_1 (1)^2 (-0.03)^1 + {}^3C_2 (1)(-0.03)^2 + {}^3C_3 (1)^0 (-0.03)^3 \\
&= 1 - .09 + .0027 + .000027 \\
&= 1.0027
\end{aligned}$$

(ii)  $(2.02)^4$

**Solution:**

$$\begin{aligned}
&(2.02)^4 \\
&= (2 + 0.02)^4
\end{aligned}$$

$$\begin{aligned}
&= {}^4C_0(2)^4(0.02)^0 + {}^4C_1(2)^3(0.02)^1 + {}^4C_2(2)^2(0.02)^2 + {}^4C_3(2)^1(0.02)^3 + {}^4C_4(2)^0(0.02)^4 \\
&= 1 \times 16 + 4(8)(0.02) + 6(4)(0.0004) + 4(2)(0.000008) + 1(0.00000016) \\
&= 16 + 0.64 + 0.0096 + 0.000064 + 0.00000016 \\
&= 16.64966416
\end{aligned}$$

(iii)  $(9.98)^4$

**Solution:**

$$\begin{aligned}
&(9.98)^4 \\
&= (10 - 0.02)^4 \\
&= {}^4C_0(10)^4(-0.02)^0 + {}^4C_1(10)^3(-0.02)^1 + {}^4C_2(10)^2(-0.02)^2 + {}^4C_3(10)^1(-0.02)^3 + {}^4C_4(10)^0(-0.02)^4 \\
&= 1 \times 10000 + 4(1000)(-0.02) + 6(100)(0.0004) - 4(10)(0.000008) + 1 \times (0.00000016) \\
&= 10000 - 80 + 0.24 + 0.00032 + 0.00000016 \\
&= 9920.228968
\end{aligned}$$

(iv)  $(21)^5$

**Solution:**

$$\begin{aligned}
&(21)^5 \\
&= (20 + 1)^5 \\
&= {}^5C_0(20)^5(1)^0 + {}^5C_1(20)^4(1)^1 + {}^5C_2(20)^3(1)^2 + {}^5C_3(20)^2(1)^3 + {}^5C_4(20)^1(1)^4 + {}^5C_5(20)^0(1)^5 \\
&= (3200000) + 5(160000) + 10(8000) + 10(400) + 5(20) + 1 \times 1 \\
&= 3200000 + 800000 + 80000 + 4000 + 100 + 1 \\
&= 4084101
\end{aligned}$$

**Q.3 Expand and simply the followings.**

(i)  $(a + \sqrt{2x})^4 + (a - \sqrt{2x})^4$

**Solution:**

$$\begin{aligned}
(a + \sqrt{2x})^4 &= {}^4C_0 a^4 (\sqrt{2x})^0 + {}^4C_1 (a^3) (\sqrt{2x})^1 + {}^4C_2 (a^2) (\sqrt{2x})^2 + {}^4C_3 (a) (\sqrt{2x})^3 \\
&\quad + {}^4C_4 (a)^0 (\sqrt{2x})^4
\end{aligned} \tag{i}$$

Similarly,

$$\begin{aligned}
(a - \sqrt{2x})^4 &= {}^4C_0 a^4 (\sqrt{2x})^0 - {}^4C_1 (a^3) (\sqrt{2x})^1 + {}^4C_2 (a^2) (\sqrt{2x})^2 \\
&\quad - {}^4C_3 (a) (\sqrt{2x})^3 + {}^4C_4 (a)^0 (\sqrt{2x})^4
\end{aligned} \tag{ii}$$

Adding (i) & (ii) we get;

$$\begin{aligned}
(a + \sqrt{2x})^4 + (a - \sqrt{2x})^4 &= 2({}^4C_0 a^4 (\sqrt{2x})^0 + {}^4C_2 (a^2) (\sqrt{2x})^2 + {}^4C_4 (a)^0 (\sqrt{2x})^4) \\
&= 2a^4 + 24a^2x^2 + 8x^4
\end{aligned}$$

(ii)  $(2 + \sqrt{3})^5 + (2 - \sqrt{3})^5$

**Solution:**

$$\begin{aligned}(2 + \sqrt{3})^5 &= {}^5C_0(2)^5(\sqrt{3})^0 + {}^5C_1(2)^4(\sqrt{3})^1 + {}^5C_2(2)^3(\sqrt{3})^2 \\ &\quad + {}^5C_3(2)^2(\sqrt{3})^3 + {}^5C_4(2)^1(\sqrt{3})^4 + {}^5C_5(2)^0(\sqrt{3})^5\end{aligned}\quad (i)$$

Similarly,

$$\begin{aligned}(2 - \sqrt{3})^5 &= {}^5C_0(2)^5(\sqrt{3})^0 - {}^5C_1(2)^4(\sqrt{3})^1 + {}^5C_2(2)^3(\sqrt{3})^2 \\ &\quad - {}^5C_3(2)^2(\sqrt{3})^3 + {}^5C_4(2)^1(\sqrt{3})^4 - {}^5C_5(2)^0(\sqrt{3})^5\end{aligned}\quad (ii)$$

Adding (i) and (ii) we get;

$$\begin{aligned}(2 + \sqrt{3})^5 + (2 - \sqrt{3})^5 &= 2({}^5C_0(2)^5(3)^0 + {}^5C_2(2)^3(\sqrt{3})^2 + {}^5C_4(2)^1(\sqrt{3})^4) \\ &= 2(32 + 10 \times 8 \times 3 + 5 \times 2 \times 9) \\ &= 64 + 480 + 180 \\ &= 724\end{aligned}$$

$$(iii) \quad (2+i)^5 - (2-i)^5$$

**Solution:**

$$\begin{aligned}(2+i)^5 &= {}^5C_0(2^5)(i)^0 + {}^5C_1(2^4)(i) + {}^5C_2(2)(i)^2 \\ &\quad + {}^5C_3(2)^2(i)^3 + {}^5C_4(2)^1(i)^4 + {}^5C_5(2)^0(i)^5\end{aligned}\quad (i)$$

Similarly;

$$\begin{aligned}(2-i)^5 &= {}^5C_0(2)^5(i)^0 - {}^5C_1(2)^4(i) + {}^5C_2(2)^3(i)^2 \\ &\quad - {}^5C_3(2)(i)^3 + {}^5C_4(2)(i)^4 + {}^5C_5(2)^0(i)^5\end{aligned}$$

Subtracting (ii) from (i)

$$\begin{aligned}(2+i)^5 - (2-i)^5 &= 2({}^5C_1(2)^4(i) + {}^5C_3(2)^2(i)^3 + {}^5C_5(2)^0(i)^5) \\ &= 2(5 \times 16i + 10 \times 4(-i) + (+i)) \\ &= 2(80i - 40i + i) \\ &= 2(41i) \\ &= 82i\end{aligned}$$

$$(iv) \quad (x + \sqrt{x^2 - 1})^3 + (x - \sqrt{x^2 - 1})^3$$

**Solution:**

$$\begin{aligned}(x + \sqrt{x^2 - 1})^3 &= {}^3C_0x^3(\sqrt{x^2 - 1})^0 + {}^3C_1(x)^2(\sqrt{x^2 - 1})^1 + {}^3C_2(x)^1(\sqrt{x^2 - 1})^2 + {}^3C_3(x)^0(\sqrt{x^2 - 1})^3\end{aligned}\quad (i)$$

Similarly,

$$(x - \sqrt{x^2 - 1})^3$$



$$= {}^3C_0 x^3 (\sqrt{x^2-1})^0 + {}^3C_1 (x)^2 (\sqrt{x^2-1})^1 + {}^3C_2 (x)^1 (\sqrt{x^2-1})^2 - {}^3C_3 (x)^0 (\sqrt{x^2-1})^3 \quad (\text{ii})$$

Adding (i) and (ii) we get;

$$\begin{aligned} (x + \sqrt{x^2-1})^3 + (x - \sqrt{x^2-1})^3 &= 2({}^3C_0 x^3 (\sqrt{x^2-1})^0 + {}^3C_2 (x^1) (\sqrt{x^2-1})^2) \\ &= 2(x^3 + 3x^1(x^2-1)) \\ &= 2(x^3 + 3x^3 - 3x) \\ &= 2(4x^3 - 3x) \\ &= 2x(4x^2 - 3) \end{aligned}$$

**Q.4 Expand the following in ascending power of x.**

(i)  $(2+x-x^2)^4$

**Solution:**

$$(2+x-x^2)^4$$

Put  $2+x=a$

then

$$\begin{aligned} (a-x^2)^4 &= {}^4C_0 (a)^4 (-x^2)^0 + {}^4C_1 (a)^3 (-x^2)^1 + {}^4C_2 (a)^2 (-x^2)^2 + {}^4C_3 (a)^1 (-x^2)^3 + {}^4C_4 (a)^0 (-x^2)^4 \\ &= a^4 - 4a^3 x^2 + 6a^2 x^4 - 4a x^6 + x^8 \end{aligned}$$

Put  $a=2+x$  back we get;

$$\begin{aligned} (2+x-x^2)^4 &= (2-x)^4 - 4x^2(2+x)^3 + 6x^4(2+x)^2 - 4x^6(2+x) + x^8 \\ &= [{}^4C_0 2^4 + {}^4C_1 (2)^3 x + {}^4C_2 (2)^2 x^2 + {}^4C_3 (2)^1 x^3 + {}^4C_4 0 x^4] \\ &\quad - 4x^2 [{}^3C_0 2^3 + {}^3C_1 (2^2)(x) + {}^3C_2 (2^1)(x^2) + {}^3C_3 x^3] + 6x^4(4+4x+x^2) - 4x^6(2+x) + x^8 \\ &= (16+32x+24x^2+8x^3+x^4) - 4x^2(8+12x+6x^2+x^3) \\ &\quad + 6x^4(4+4x+x^2) - 4x^6(2+x) + x^8 \\ &= 16+32x-8x^2-40x^3+x^4+20x^5-2x^6-4x^7+x^8 \end{aligned}$$

(ii)  $(1-x+x^2)^4$

**Solution:**

Let  $(1-x)=a$  then

$$(1-x+x^2)^4 = (a+x^2)^4$$

$$(1-x+x^2)^4 = (a+x^2)^4$$

$$\begin{aligned} &= {}^4C_0 (a)^4 + {}^4C_1 (a)^3 (x^2)^1 + {}^4C_2 (a)^2 (x^2)^2 + {}^4C_3 (a)^1 (x^2)^3 + {}^4C_4 (a)^0 (x^2)^4 \\ &= a^4 + 4a^3 x^2 + 6a^2 x^4 + 4a x^6 + x^8 \end{aligned}$$

Put  $a=1-x$  Back, we get;

$$= (1-x)^4 + 4x^2(1-x)^3 + 6x^4(1-x)^2 + 4x^6(1-x) + 8x^8$$

$$\begin{aligned}
&= 1 - 4x + 6x^2 - 4x^3 + x^4 + 4x^2 - 12x^3 + 12x^4 - 4x^5 + 6x^4 - 12x^5 + 6x^6 + 4x^6 - 4x^5 + x^8 - 16x^5 \\
&= \{1 - 4x + 6x^2 - 4x^3 + x^4\} + 4x^2\{1 - 3x + 3x^2 - x^3\} + 6x^4(1 - 2x + x^2) + 4x^6(1 - x) + x^8 \\
&= 1 - 4x + 6x^2 + 4x^2 - 4x^3 - 12x^3 + x^4 + 12x^4 + 6x^4 - 4x^5 - 12x^5 + 6x^6 + 4x^6 - 4x^7 + x^8 \\
&= 1 - 4x + 10x^2 - 16x^3 + 19x^4 - 16x^5 + 10x^6 - 4x^7 + x^8
\end{aligned}$$

(iii)  $(1 - x - x^2)^4$

**Solution:**

Let  $(1 - x) = a$  then

$$\begin{aligned}
(1 - x - x^2)^4 &= (a - x^2)^4 \\
&= a^4 - 4a^3x^2 + 6a^2x^4 - 4a^1x^6 + x^8
\end{aligned}$$

Put back  $a = 1 - x$  we get;

$$\begin{aligned}
&= (1 - x)^4 + 4x^2(1 - x)^3 + 6x^4(1 - x)^2 - 4x^6(1 - x) + x^8 \\
&= \left(1 + \binom{4}{1}1 \cdot (-x) + \binom{4}{2}1 \cdot (-x)^2 + \binom{4}{3}1 \cdot (-x)^3 + \binom{4}{4}1 \cdot (-x)^4\right) \\
&\quad + 4x^2\left(1 + \binom{3}{1}1 \cdot (-x) + \binom{3}{2}1 \cdot (-x)^2 + \binom{3}{3}1 \cdot (-x)^3\right) + 6x^4(1 - 2x + x^2) - 4x^6 + 4x^7 + x^8 \\
&= 1 - 4x + 6x^2 - 4x^3 + x^4 + 4x^2 - 12x^3 + 12x^4 - 4x^5 + 6x^4 - 2x^5 + 6x^6 + 4x^6 - 4x^7 + x^8 \\
&= 1 - 4x + 10x^2 - 16x^3 + 19x^4 - 16x^5 + 10x^6 - 4x^7 + x^8
\end{aligned}$$

**Q.5 Expand the following in descending powers of  $x$ .**

(i)  $(x^2 + x - 1)^3$

**Solution:**

Let  $x - 1 = a$  then

$$\begin{aligned}
(x^2 + x - 1)^3 &= (x^2 + a)^3 \\
&= {}^3C_0(x^2)^3(a)^0 + {}^3C_1(x^2)^2(a)^1 + {}^3C_2(x^2)^1(a)^2 + {}^3C_3(x^2)^0a^3
\end{aligned}$$

pulling back  $x - 1 = a$  we get;

$$\begin{aligned}
&= x^6 + 3x^4(x - 1) + 3x^2(x - 1)^2 + (x - 1)^3 \\
&= x^6 + 3x^5 - 3x^4 + 3x^2(x^2 - 2x + 1) + (x^3 - 3x^2 + 3x - 1) \\
&= x^6 + 3x^5 - 3x^4 + 3x^4 - 6x^3 + 3x^2 + x^3 - 3x^2 + 3x - 1 \\
&= x^6 + 3x^5 - 5x^3 + 3x - 1
\end{aligned}$$

(ii)  $\left(x - 1 - \frac{1}{x}\right)^3$

**Solution:**

Let  $x - 1 = a$  then

$$\left(x - 1 - \frac{1}{x}\right)^3 = \left(a - \frac{1}{x}\right)^3$$

$$\begin{aligned}
&= {}^3C_0(a)^3\left(\frac{-1}{x}\right)^0 + {}^3C_1(a)^2\left(\frac{-1}{x}\right) + {}^3C_2(a)^1\left(\frac{-1}{x}\right)^2 + {}^3C_3(a)^0\left(\frac{-1}{x}\right)^3 \\
&= a^3 - 3a^2\left(\frac{1}{x}\right) + 3a\left(\frac{1}{x^2}\right) - \frac{1}{x^3}
\end{aligned}$$

Put a  $a = x - 1$  back we get;

$$\begin{aligned}
&= (x-1)^3 - 3\left(\frac{1}{x}\right)(x-1)^2 + 3\left(\frac{1}{x^2}\right)(x-1) - \frac{1}{x^3} \\
&= (x^3 - 3x^2 + 3x - 1) - \frac{3}{x}(x^2 - 2x + 1) + \frac{3}{x^2}(x-1) - \frac{1}{x^3} \\
&= x^3 - 3x^2 + \cancel{3x} - 1 - \cancel{3x} + \frac{\cancel{-3}}{x} + \frac{\cancel{3}}{x} - \frac{3}{x^2} - \frac{1}{x^3} \\
&= x^3 - 3x^2 + 5 - \frac{3}{x^2} - \frac{1}{x^3}
\end{aligned}$$

**Q.6 Find the term involving:**

**(i)  $x^4$  in the expansion of  $(3 - 2x)^7$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $n = 7, a = 3, b = 2x$  so we get;

$$T_{r+1} = {}^7C_r (3)^{7-r} (-2x)^r$$

For the term involving  $x^4$ , put exponent of  $x$  equal to 4 we get  $r = 4$

So

$$T_{4+1} = {}^7C_4 (3)^{7-4} (-2x)^4$$

$$T_5 = 35(27)(16)x^4$$

$$T_5 = 15120x^4$$

**(ii)  $x^{-2}$  in the expansion of  $\left(x - \frac{2}{x^2}\right)^{13}$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $n = 13, a = x, b = \frac{-2}{x^2}$  we get;

$$T_{r+1} = {}^{13}C_r (x)^{13-r} \left(\frac{-2}{x^2}\right)^r$$

$$T_{r+1} = {}^{13}C_r x^{13-r} (-2)^r x^{-2r}$$

$$= {}^{13}C_r (-2)^r x^{13-3r}$$

For the term involving  $x^{-2}$  put the exponent of  $x$  equal to  $-2$  we get;

$$13 - 3r = -2 \Rightarrow 15 = 3r$$

$r = 5$  we get;

$$\begin{aligned} T_{5+1} &= {}^{13}C_5 (-2)^5 (x)^{-2} \\ &= (1287)(-32)x^{-2} \end{aligned}$$

$$T_6 = \frac{-41184}{x^2}$$

**(iii)  $a^4$  in the expansion of  $\left(\frac{2}{x} - a\right)^9$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $a = \frac{2}{x}, b = -a, n = 9$  we get;

$$T_{r+1} = {}^9C_r \left(\frac{2}{x}\right)^{9-r} (-a)^r$$

For the term involving  $a^4$  put exponent of  $a$  equal to 4 i.e.,  $r = 4$

So

$$T_{4+1} = {}^9C_4 \left(\frac{2}{x}\right)^5 (-a)^4$$

$$T_5 = \frac{4032a^4}{x^5}$$

**(iv)  $y^3$  in the expansion of  $(x - \sqrt{y})^{11}$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $n = 11, a = x, b = -\sqrt{y}$

So

$$T_{r+1} = {}^{11}C_r (x)^{11-r} (-\sqrt{y})^r, \text{ Suppose } y \text{ occurs in } T_{r+1} \text{ i.e.,}$$

$$y^3 = y^{\frac{r}{2}} \Rightarrow \frac{r}{2} = 3$$

$$r = 6$$

Now,

$$T_{6+1} = {}^{11}C_6 (x)^{11-6} (-y^{r/2})^6$$

$$T_{6+1} = \binom{11}{6} x^5 y^3$$

$$T_7 = 462 x^5 y^3$$

**Q.7 Find the coefficient of;**

**(i)  $x^5$  in the expansion of  $\left(x^2 - \frac{3}{2x}\right)^{10}$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $a = x, b = -\frac{3}{2x}, n = 10$

So

$$\begin{aligned} T_{r+1} &= {}^{10}C_r (x^2)^{10-r} \left(\frac{-3}{2x}\right)^r \\ &= {}^{10}C_r x^{20-2r} \left(\frac{-3}{2}\right)^r x^{-r} \\ &= {}^{10}C_r \left(\frac{-3}{2}\right)^r x^{20-2r-r} \\ &= {}^{10}C_2 \left(-\frac{3}{2}\right)^r (x)^{20-3r} \end{aligned}$$

For the term involving  $x^5$ , put  $20-3r=5$  we get; i.e.,

$$20-3r=5$$

$$15=3r$$

$$5=r$$

$$T_{5+1} = {}^{10}C_5 \left(\frac{-3}{2}\right)^5 x^5$$

$$T_6 = \frac{-15309}{8} x^5$$

Thus coefficient of  $x^5$  is  $\frac{-15309}{8} = -1913.625$

**(ii)  $x^n$  in the expansion of  $\left(x^2 - \frac{1}{x}\right)^{2n}$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $n = 2n, a = x^2, b = -\frac{1}{x}$  we get;

$$\begin{aligned} T_{r+1} &= {}^{2n}C_r (x^2)^{2n-r} \left(-\frac{1}{x}\right)^r \\ &= {}^{2n}C_r (x^2)^{4n-2r} (-1)^r x^{-r} \\ &= {}^{2n}C_r x^{4n-3r} (-1)^r \end{aligned}$$

For the term involving  $x^n$  put the exponent of  $x$  equal to  $n$ , so

$$4n - 3r = n$$

$$3n = 3r$$

$$r = n$$

Thus

$$T_{n+1} = {}^{2n}C_n x^n (-1)^n$$

$$T_{n+1} = \frac{(2n)!}{n! \times n!} (-1)^n x^n$$

So the coefficient of  $x^n$  is  $\frac{(-1)^n (2n)!}{(n!)^2}$

**Q.8 Find the 6<sup>th</sup> term in the expansion of  $\left(x^2 - \frac{3}{2x}\right)^{10}$**

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $a = x^2, b = -\frac{3}{2x}, n = 10$

So, we get

$$T_{r+1} = {}^{10}C_r (x^2)^{10-r} \left(-\frac{3}{2x}\right)^r$$

for the 6<sup>th</sup> term put  $r = 5$  we get;

$$T_{5+1} = {}^{10}C_5 (x^2)^{10-5} \left(-\frac{3}{2x}\right)^5$$

$$= 252x^{10} \times \left(-\frac{243}{32}\right) \times \frac{1}{x^5}$$

$$T_6 = \frac{-15309}{8} x^5 = -1913.625x^5$$

**Q.9 Find the term independent of  $x$  in the following expansions.**

(i)  $\left(x - \frac{2}{x}\right)^{10}$

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $n=10, a=x, b=\frac{-2}{x}$

$$\begin{aligned} T_{r+1} &= {}^{10}C_r (x)^{10-r} \left(\frac{-2}{x}\right)^r \\ &= {}^{10}C_r (x^{10-r}) (x^{-r}) (-2)^r \\ &= {}^{10}C_r x^{10-2r} (-2)^r \end{aligned}$$

For the term involving  $x^0$  (term independent from  $x$ ) put exponent of  $x$  equal to zero i.e.,

$$10-2r=0$$

$$r=5$$

Thus

$$\begin{aligned} T_{5+1} &= {}^{10}C_5 x^0 (-2)^5 \\ &= \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} (-32) \end{aligned}$$

$$T_6 = -8064$$

$$(ii) \quad \left(\sqrt{x} + \frac{1}{2x^2}\right)^{10}$$

**Solution:**

As we know that  $(r+1)^{th}$  term in the expansion of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $n=10, a=\sqrt{x}, b=\frac{1}{2x^2}$

$$\begin{aligned} T_{r+1} &= {}^{10}C_r (\sqrt{x})^{10-r} \left(\frac{1}{2x^2}\right)^r \\ &= {}^{10}C_r (x)^{5-\frac{r}{2}} \left(\frac{1}{2}\right)^r x^{-2r} \\ &= {}^{10}C_r \left(\frac{1}{2}\right)^r x^{5-\frac{r}{2}-2r} \quad \because 5-\frac{r}{2}-2r \\ T_{r+1} &= {}^{10}C_r \left(\frac{1}{2}\right)^r x^{5-\frac{5r}{2}} \quad \because \frac{10-r-4r}{2} = \frac{10-5r}{2} = 5-\frac{5}{2}r \end{aligned}$$

For the term involving  $x^0$  (term independent from  $x$ ) put exponent of  $x$  equal to zero i.e.,

$$\text{Put } 5-\frac{5r}{2}=0$$

$$\frac{5r}{2}=5$$

$$r=2$$

Put  $r = 2$  in the last expansion we get;

$$T_3 = {}^{10}C_2 \left(\frac{1}{2}\right)^2 x^0$$

$$= \frac{10 \times 9}{2} \times \frac{1}{4}$$

$$T_3 = \frac{45}{4}$$

$$\text{(iii)} \quad (1+x^2)^3 \left(1+\frac{1}{x^2}\right)^4$$

**Solution:**

$$(1+x^2)^3 \left(\frac{x^2+1}{x^2}\right)^4$$

$$= (1+x^2)^3 \frac{(1+x^2)^4}{x^8}$$

$$= x^{-8} (1+x^2)^7$$

$(r+1)^{\text{th}}$  term in the expansion of  $(1+x^2)^7$  is

$$T_{r+1} = {}^7C_r (1)^{7-r} (x^2)^r$$

$$= {}^7C_r x^{2r}$$

Thus

$$= x^{-8} \times {}^7C_r x^{2r}$$

$$= {}^7C_r x^{2r-8}$$

For the term involving  $x^0$  (term independent from  $x$ ) put exponent of  $x$  equal to zero i.e.,

$$\text{Put } 2r-8=0 \Rightarrow r=4$$

So required term independent from  $x$  is

$$T_{4+1} = {}^7C_4 x^0$$

$$= 35$$

**Q.10 Determine the middle term in the following expansions:**

$$\text{(i)} \quad \left(\frac{1}{x} - \frac{x^2}{2}\right)^{12}$$

**Solution:**

Here  $n=12$  (Even) so that middle term is  $\left(\frac{n}{2}+1\right)^{\text{th}}$  term i.e.,

$$\left(\frac{n}{2}+1\right)^{\text{th}} = \left(\frac{12}{2}+1\right)^{\text{th}} = 7^{\text{th}} \text{ term}$$

$$\text{Here } a = \frac{1}{x}, b = \frac{-x^2}{2}, n=12, r=6$$



So

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

$$\begin{aligned} T_{6+1} &= {}^{12}C_6 \left(\frac{1}{x}\right)^6 \left(\frac{-x^2}{2}\right)^6 \\ &= 924 \left(\frac{1}{x^6}\right) \left(\frac{x^{12}}{64}\right) \end{aligned}$$

$$T_7 = \frac{231}{16} x^6$$

$$(ii) \quad \left(\frac{3}{2}x - \frac{1}{3x}\right)^{11}$$

**Solution:**

Here  $n=11$  (odd), so the middle terms are  $\left(\frac{n+1}{2}\right)^{th}$  and  $\left(\frac{n+3}{2}\right)^{th}$

So

$$\left(\frac{n+1}{2}\right)^{th} = \left(\frac{11+1}{2}\right)^{th} = 6^{th} \text{ term}$$

$$\left(\frac{n+3}{2}\right)^{th} = \left(\frac{11+3}{2}\right)^{th} = 7^{th} \text{ term}$$

Here  $a = \frac{3}{2}, b = -\frac{1}{3x}, n = 11$

**For 6<sup>th</sup> term:**

$$r = 5$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

$$T_6 = {}^{11}C_5 \left(\frac{3}{2}x\right)^6 \left(\frac{-1}{3x}\right)^5$$

$$T_6 = \frac{-693}{32} x$$

**For 7<sup>th</sup> term:**

$$r = 6$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

$$\begin{aligned} T_7 &= {}^{11}C_6 \left(\frac{3}{2}x\right)^5 \left(-\frac{1}{3x}\right)^6 \\ &= 462 \left(\frac{3^5 x^5}{2^5}\right) \left(\frac{1}{3^6 x^6}\right) \\ &= 462 \left(\frac{1}{32 \times 3x}\right) \end{aligned}$$

$$(iii) \quad \left(2x - \frac{1}{2x}\right)^{2m+1}$$

**Solution:**

Here  $2m+1 = \text{odd}$  so the middle terms are

$$\left(\frac{n+1}{2}\right)^{th} \text{ and } \left(\frac{n+3}{2}\right)^{th} \text{ terms}$$

So

$$\left(\frac{n+1}{2}\right)^{th} = \left(\frac{2m+1+1}{2}\right)^{th} = (m+1)^{th} \text{ term}$$

$$\left(\frac{n+3}{2}\right)^{th} = \left(\frac{2m+1+3}{2}\right)^{th} = (m+2)^{th} \text{ term}$$

$$\text{Here } a = 2x, b = -\frac{1}{2x}, n = 2m+1, r = m$$

**For  $(m+1)^{th}$  term:**

$$r = m$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

$$\begin{aligned} T_{m+1} &= {}^nC_m (2x)^{n-m} \left(\frac{-1}{2x}\right)^m \\ &= {}^{2m+1}C_m (2x)^{2m+1-m} \left(\frac{-1}{2x}\right)^m \\ &= \frac{(2m+1)!}{m!(m+1)!} (2x)^{m+1} (-1)^m \times \frac{1}{(2x)^m} \\ &= \frac{(2m+1)!}{m!(m+1)!} (-1)^m 2x \end{aligned}$$

**For  $(m+2)^{th}$  term:**

$$r = m+1$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

$$\begin{aligned} T_{m+2} &= {}^nC_{m+1} (2x)^{n-(m+1)} \left(\frac{-1}{2x}\right)^{m+1} \\ T_{m+2} &= {}^{2m+1}C_{m+1} (2x)^{2m+1-m-1} \left(\frac{-1}{2x}\right)^{m+1} \\ &= \frac{(2m+1)!}{(m+1)!(2m+1-m-1)!} (2x)^m \left(\frac{-1}{2x}\right)^{m+1} \\ &= \frac{(2m+1)!}{(m+1)!(m)!} (-1)^{m+1} \cdot \frac{(2x)^m}{(2x)^{m+1}} \end{aligned}$$

$$= \frac{(2m+1)!}{m!(m+1)!} (-1)^{m+1} \frac{1}{2x}$$

**Q.11** Find  $(2n+1)^{\text{th}}$  term from the end in the expansion of  $\left(x - \frac{1}{2x}\right)^{3n}$

**Solution:**

$(2n+1)^{\text{th}}$  term from the end in the expansion of  $\left(x - \frac{1}{2x}\right)^{3n}$  is  $(2n+1)^{\text{th}}$  term from the beginning in the expansion of  $\left(\frac{-1}{2x} + x\right)^{3n}$

As  $(r+1)^{\text{th}}$  term of  $(a+b)^n$  is

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Put  $r = 2n, a = \frac{-1}{2x}, b = x, n = 3n$  we get;

$$\begin{aligned} T_{2n+1} &= {}^{3n}C_{2n} \left(\frac{-1}{2x}\right)^{3n-2n} (x)^{2n} \\ &= \frac{(3n)!}{(3n-2n)! \times (2n)!} \times \left(\frac{-1}{2}\right)^n \times \frac{1}{x^n} \times x^{2n} \\ T_{2n+1} &= \frac{(-1)^n}{2^n} \times \frac{(3n)!}{(2n)! \times n!} x^n \end{aligned}$$

**Q.12** Show that middle term of  $(1+x)^{2n}$  is  $\frac{1.3.5 \dots (2n-1)}{n!} 2^n x^n$

**Proof:**

Here  $2n = \text{even}$  so the middle term is  $\left(\frac{n}{2} + 1\right)^{\text{th}} = \left(\frac{2n}{2} + 1\right)^{\text{th}} = (n+1)^{\text{th}}$  term.

Thus

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Here  $r = n, a = 1, b = x, n = 2n$

So

$$\begin{aligned} T_{n+1} &= {}^{2n}C_n (1)^{2n-n} (x)^n \\ &= \frac{(2n)!}{(2n-n)! \times n!} x^n \\ &= \frac{(2n)!}{n! \times n!} x^n \\ &= \frac{(2n)(2n-1)(2n-2)(2n-3)(2n-4) \dots 4.3.2.1}{n! \times n!} x^n \end{aligned}$$

$$\begin{aligned}
&= \frac{[(2n)(2n-2)(2n-4).....4.2][(2n-1)(2n-3)2n-5.....3.1]}{n! \times n!} \times x^n \\
&= \frac{2^n \{n(n-1)(n-2).....2.1\} \{(2n-1)(2n-3)(2n-5).....3.1\}}{n! \times n!} \times x^n \\
&= \frac{2^n \times n! \times \{(2n-1)(2n-3)(2n-5).....3.1\}}{n! \times n!} x^n \\
&= \frac{2^n \times \{(2n-1)(2n-3)(2n-5).....5.3.1\}}{n!} x^n \\
&= \frac{\{1.3.5....(2n-1)\} 2^n x^n}{n!} \\
&= \frac{1.3.5....(2n-1)}{n!} 2^n x^n
\end{aligned}$$

Hence the proof.

**Q.13 Show that**

$$\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + ..... \binom{n}{n-1} = 2^{n-1}$$

**Proof:**

We know that

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + ..... + \binom{n}{n}x^n \quad (i)$$

Put  $x = 1$  in (i) we get;

$$\begin{aligned}
(1+1)^n &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ..... + \binom{n}{n} \\
2^n &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ..... + \binom{n}{n} \quad (ii)
\end{aligned}$$

Put  $x = -1$  in (i) we get;

$$\begin{aligned}
(1+(-1))^n &= \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + ..... + (-1)^{n-1} \binom{n}{n-1} + (-1)^n \binom{n}{n} \\
0 &= \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + ..... + (-1)^{n-1} \binom{n}{n-1} + (-1)^n \binom{n}{n}
\end{aligned}$$

Assume that here  $n$  is even.

$$\begin{aligned}
\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + ... + \binom{n}{n} &= \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + ... + \binom{n}{n-1} \quad (iii) \\
2^n &= \left\{ \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + ..... + \binom{n}{n} \right\} + \left\{ \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + ..... + \binom{n}{n-1} \right\}
\end{aligned}$$

Using (iii) we get;

$$2^n = \left\{ \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + ... + \binom{n}{n-1} \right\} + \left\{ \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + ... + \binom{n}{n-1} \right\}$$

$$2^n = 2 \left\{ \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots + \binom{n}{n-1} \right\}$$

$$\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots + \binom{n}{n-1} = 2^{n-1}$$

Hence the proof

**Q.14** Show that

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \frac{1}{4}\binom{n}{3} + \dots + \frac{1}{n+1}\binom{n}{n} = \frac{2^{n+1}-1}{n+1}$$

**Proof:**

$$\begin{aligned} \text{L.H.S} &= \binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \frac{1}{4}\binom{n}{3} + \dots + \frac{1}{n+1}\binom{n}{n} \\ &= \frac{n!}{0!(n-0)!} + \frac{1}{2} \times \frac{n!}{1!(n-1)!} + \frac{1}{3} \frac{n!}{2!(n-2)!} + \frac{1}{4} \frac{n!}{3!(n-3)!} + \dots + \frac{1}{n+1} \frac{n!}{n!(n-n)!} \\ &= 1 + \frac{n}{2!} + \frac{n(n-1)}{3!} + \frac{n(n-1)(n-2)}{4!} + \dots + \frac{1}{(n+1)} \end{aligned}$$

Taking common  $\frac{1}{n+1}$  we get;

$$= \frac{1}{n+1} \left[ (n+1) + \frac{(n+1)n}{2!} + \frac{(n+1)n(n-1)}{3!} + \frac{(n+1)n(n-1)(n-2)}{4!} + \dots + 1 \right]$$

Above expression can be written as;

$$\begin{aligned} &= \frac{1}{n+1} \left[ \binom{n+1}{1} + \binom{n+1}{2} + \binom{n+1}{3} + \dots + \binom{n+1}{n+1} \right] \\ &= \frac{1}{n+1} \left[ \binom{n+1}{0} + \binom{n+1}{1} + \binom{n+1}{2} + \binom{n+1}{3} + \dots + \binom{n+1}{n+1} - \binom{n+1}{0} \right] \\ &= \frac{1}{n+1} [2^{n+1} - 1] \\ &= \text{R.H.S} \end{aligned}$$

**The Binomial Theorem when the index n is a negative integer or a fraction.**

When n is negative integer or a fraction, then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r + \dots$$

Provide  $|x| < 1$

The series of the type

$$1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

Is called that binomial series

**Note:**

- (1) The proof of this theorem is beyond the scope of this book.
- (2) Symbols  $\binom{n}{0}\binom{n}{1}\binom{n}{2}$  etc are meaningless when n is negative integer or a fraction.
- (3) The general term in the expansion is  $T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r$

**Some particular cases of the expansion of  $(1+x)^n$ ,  $n < 0$** 

- (i)  $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r + \dots$
- (ii)  $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots + (-1)^r (r+1)x^r + \dots$
- (iii)  $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots + (-1)^r \frac{(r+1)(r+2)}{2}x^r + \dots$
- (iv)  $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r + \dots$
- (v)  $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots + (r+1)x^r + \dots$
- (vi)  $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots + \frac{(r+1)(r+2)}{2}x^r + \dots$

**EXERCISE 8.3**

**Q.1** Expand the following upto 4 terms, taking the value of  $x$  such that the expansion in each case is valid.

(i)  $(1-x)^{\frac{1}{2}}$

**Solution:**

$$\begin{aligned}(1-x)^{\frac{1}{2}} &= 1 + \left(\frac{1}{2}\right)(-x) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!}(-x)^2 + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}(-x)^3 + \dots \\&= 1 - \frac{1}{2}x + \left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)\frac{1}{2}x^2 + \frac{1}{2} \times \left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)\frac{1}{6}(-x^3) \dots \\&= 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3 \dots\end{aligned}$$

The expansion of  $(1-x)^{\frac{1}{2}}$  is valid if  $|x| < 1$

(ii)  $(1+2x)^{-1}$

**Solution:**

$$\begin{aligned}(1+2x)^{-1} &= 1 + (-1)(2x) + \frac{(-1)(-1-1)}{2!}(2x)^2 + \frac{(-1)(-1-1)(-1-2)}{3!}(2x)^3 + \dots \\&= 1 - 2x + 4x^2 - 8x^3 + \dots\end{aligned}$$

The expansion of  $(1+2x)^{-1}$  is valid if  $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$

(iii)  $(1+x)^{-\frac{1}{3}}$

**Solution:**

$$\begin{aligned}(1+x)^{-\frac{1}{3}} &= 1 + \left(\frac{-1}{3}\right)x + \frac{\left(\frac{-1}{3}\right)\left(\frac{-1}{3}-1\right)}{2!}x^2 + \frac{\left(\frac{-1}{3}\right)\left(\frac{-1}{3}-1\right)\left(\frac{-1}{3}-2\right)}{3!}x^3 + \dots \\&= 1 - \frac{1}{3}x + \frac{\left(\frac{-1}{3}\right)\left(\frac{-4}{3}\right)}{2}x^2 + \frac{\left(\frac{-1}{3}\right)\left(\frac{-4}{3}\right)\left(\frac{-7}{3}\right)}{6}x^3 + \dots \\&= 1 - \frac{1}{3}x + \frac{2}{9}x^2 - \frac{4 \times 7}{27 \times 3 \times 2}x^3 + \dots \\&= 1 - \frac{1}{3}x + \frac{2}{9}x^2 - \frac{14}{81}x^3 + \dots\end{aligned}$$

The expansion of  $(1+x)^{-\frac{1}{3}}$  is valid if  $|x| < 1$

(iv)  $(4-3x)^{\frac{1}{2}}$

**Solution:**

$$(4-3x)^{\frac{1}{2}} = \left[4\left(1-\frac{3}{4}x\right)\right]^{\frac{1}{2}}$$



$$\begin{aligned}
&= 4^{\frac{1}{2}} \left[ 1 - \frac{3}{4}x \right]^{\frac{1}{2}} \\
&= 2 \left[ 1 + \frac{1}{2} \left( \frac{-3}{4}x \right) + \frac{\frac{1}{2} \left( \frac{1}{2} - 1 \right)}{2!} \left( \frac{-3}{4}x \right)^2 + \frac{\left( \frac{1}{2} \right) \left( \frac{1}{2} - 1 \right) \left( \frac{1}{2} - 2 \right)}{3!} \left( \frac{-3}{4}x \right)^3 + \dots \right] \\
&= 2 \left[ 1 - \frac{3}{8}x + \frac{\left( \frac{1}{2} \right) \left( \frac{-1}{2} \right)}{2} \frac{9x^2}{16} + \frac{\left( \frac{1}{2} \right) \left( \frac{-1}{2} \right) \left( \frac{-3}{2} \right)}{6} \left( \frac{-27}{64}x^3 \right) + \dots \right] \\
&= 2 \left[ 1 - \frac{3}{8}x - \frac{1}{8} \times \frac{9}{16}x^2 - \frac{1}{16} \times \frac{27}{64}x^3 + \dots \right] \\
&= 2 - \frac{3}{4}x - \frac{9}{64}x^2 - \frac{27}{512}x^3 + \dots
\end{aligned}$$

The expansion of  $(4-3x)^{\frac{1}{2}}$  is valid if  $\left| \frac{3}{4}x \right| < 1 \Rightarrow |x| < \frac{4}{3}$

(v)  $(8-2x)^{-1}$

**Solution:**

$$\begin{aligned}
(8-2x)^{-1} &= 8^{-1} \left( 1 - \frac{x}{4} \right)^{-1} \\
&= \frac{1}{8} \left[ 1 + (-1) \left( \frac{-x}{4} \right) + \frac{(-1)(-1-1)}{2!} \left( \frac{-x}{4} \right)^2 + \frac{(-1)(-1-1)(-1-2)}{3!} \left( \frac{-x}{4} \right)^3 + \dots \right] \\
&= \frac{1}{8} \left[ 1 + \frac{x}{4} + \frac{1 \times 2}{2 \times 1} \left( \frac{x^2}{16} \right) - \frac{2 \times 3}{6} \left( \frac{-x^3}{64} \right) + \dots \right] \\
&= \frac{1}{8} \left[ 1 + \frac{x}{4} + \frac{x^2}{16} + \frac{x^3}{64} + \dots \right] \\
&= \frac{1}{8} + \frac{1}{32}x + \frac{1}{128}x^2 + \frac{1}{512}x^3 + \dots
\end{aligned}$$

The expansion of  $(8-2x)^{-1}$  is valid if  $\left| \frac{x}{4} \right| < 1 \Rightarrow |x| < 4$

(vi)  $(2-3x)^{-2}$

**Solution:**

$$\begin{aligned}
(2-3x)^{-2} &= \left[ 2 \left( 1 - \frac{3}{2}x \right) \right]^{-2} \\
&= (2)^{-2} \left( 1 - \frac{3}{2}x \right)^{-2} \\
&= \frac{1}{4} \left[ 1 + (-2) \left( \frac{-3}{2}x \right) + \frac{(-2)(-2-1)}{2!} \left( \frac{-3}{2}x \right)^2 + \frac{(-2)(-2-1)(-2-2)}{3!} \left( \frac{-3}{2}x \right)^3 + \dots \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \left[ 1 + 3x + \frac{(-2)(-3)}{2} \left( \frac{9}{4} x^2 \right) + \frac{(-2)(-3)(-4)}{6} \left( \frac{-27}{8} x^3 \right) + \dots \right] \\
&= \frac{1}{4} \left[ 1 + 3x + \frac{27}{4} x^2 + \frac{27}{2} x^3 + \dots \right] \\
&= \frac{1}{4} + \frac{3}{4} x + \frac{27}{16} x^2 + \frac{27}{8} x^3 + \dots
\end{aligned}$$

The expansion of  $(2-3x)^{-2}$  is valid if  $\left| \frac{3}{2}x \right| < 1 \Rightarrow |x| < \frac{2}{3}$

(vii)  $\frac{(1-x)^{-1}}{(1+x)^2}$

**Solution:**

$$\begin{aligned}
\frac{(1-x)^{-1}}{(1+x)^2} &= (1-x)^{-1} (1+x)^{-2} \\
&= \left[ (1-x)^{-1} (1+x)^{-2} \right] \\
&= \left[ 1 + (-1)(-x) + \frac{(-1)(-1-1)}{2!} (-x)^2 + \frac{(-1)(-1-1)(-1-2)}{3!} (-x)^3 + \dots \right] \\
&\times \left[ 1 + (-2)(x) + \frac{(-2)(-2-1)}{2!} x^2 + \frac{(-2)(-2-1)(-1-2)}{3!} x^3 + \dots \right] \\
&= \left[ 1 + x + x^2 + x^3 + \dots \right] \left[ 1 - 2x + 3x^2 - 4x^3 + \dots \right] \\
&= 1 - 2x + 3x^2 - 4x^3 + x - 2x^2 + 3x^3 - 4x^3 + \dots \\
&\text{Neglect } x^4 \text{ and higher powers of } x \\
&= 1 - x + 2x^2 - 2x^3 + \dots
\end{aligned}$$

The expansion of  $(1-x)^{-1}$  and  $(1+x)^{-2}$  are valid if  $|x| < 1$

(viii)  $\frac{\sqrt{1+2x}}{1-x}$

**Solution:**

$$\begin{aligned}
\frac{\sqrt{1+2x}}{1-x} &= (1+2x)^{\frac{1}{2}} (1-x)^{-1} \\
&= \left[ 1 + \left( \frac{1}{2} \right) (2x) + \frac{\left( \frac{1}{2} \right) \left( \frac{1}{2} - 1 \right)}{2!} (2x)^2 + \frac{\left( \frac{1}{2} \right) \left( \frac{1}{2} - 1 \right) \left( \frac{1}{2} - 2 \right)}{3!} (2x)^3 + \dots \right] \\
&\times \left[ 1 + (-1)(-x) + \frac{(-1)(-1-1)}{2!} x^2 + \frac{(-1)(-1-1)(-1-2)}{3!} (-x)^3 + \dots \right]
\end{aligned}$$

$$\begin{aligned}
&= \left[ 1+x + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2} 4x^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{6} 8x^3 + \dots \right] \times [1+x+x^2+x^3+\dots] \\
&= \left( 1+x - \frac{1}{2}x^2 + \frac{1}{2}x^3 + \dots \right) (1+x+x^2+x^3+\dots) \\
&= 1+x+x^2+x^3+x+x^3+x^3 - \frac{1}{2}x^2 - \frac{1}{2}x^3 + \frac{1}{2}x^3 + \dots \\
&= 1+2x+\frac{3}{2}x^2+2x^3+\dots
\end{aligned}$$

The expansion of  $(1+2x)^{\frac{1}{2}}$  is valid  $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$

and the expansion of  $(1-x)^{\frac{1}{2}}$  is valid if  $|x| < 1$

So, the expansion of  $\frac{\sqrt{1+2x}}{1-x}$  is valid if  $|x| < \frac{1}{2}$

(ix)  $\frac{(4+2x)^{\frac{1}{2}}}{2-x}$

**Solution:**

$$\begin{aligned}
\frac{(4+2x)^{\frac{1}{2}}}{2-x} &= \frac{\left(4\left(1+\frac{2}{4}x\right)\right)^{\frac{1}{2}}}{2\left(1-\frac{x}{2}\right)} = \frac{4^{\frac{1}{2}}\left(1+\frac{x}{2}\right)^{\frac{1}{2}}\left(1-\frac{x}{2}\right)^{-1}}{2\left(1-\frac{x}{2}\right)} \\
&= \left(1+\frac{x}{2}\right)^{\frac{1}{2}}\left(1-\frac{x}{2}\right)^{-1} \\
&= \left[ 1 + \frac{1}{2}\left(\frac{x}{2}\right) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!}\left(\frac{x}{2}\right)^2 + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}\left(\frac{x}{2}\right)^3 + \dots \right] \\
&\times \left[ 1 + (-1)\left(-\frac{x}{2}\right) + \frac{(-1)(-1-1)}{2!}\left(-\frac{x}{2}\right)^2 + \frac{(-1)(-1-1)(-1-2)}{3!}\left(-\frac{x}{2}\right)^3 + \dots \right] \\
&= \left[ 1 + \frac{x}{4} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{6}\frac{x^3}{8} + \dots \right] \times \left[ 1 + \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{8} + \dots \right] \\
&= \left[ 1 + \frac{x}{4} - \frac{1}{32}x^2 + \frac{1}{128}x^3 \right] \left[ 1 + \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{8} + \dots \right]
\end{aligned}$$

$$\begin{aligned}
&= \left(1 + \frac{x}{4} - \frac{1}{32}x^2 + \frac{1}{128}x^3 + \dots\right) \left(1 + \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{8} + \dots\right) \\
&= 1 + \frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{8} + \frac{x}{4} + \frac{x^2}{8} + \frac{x^3}{16} - \frac{1}{32}x^2 - \frac{1}{64}x^3 + \frac{1}{128}x^3 + \dots \\
&= 1 + \frac{3}{4}x + \frac{11}{32}x^2 + \frac{23}{128}x^3 + \dots
\end{aligned}$$

The expansion of  $\left(1 + \frac{x}{2}\right)^{\frac{1}{2}}$  and  $\left(1 - \frac{x}{2}\right)^{-1}$  is valid if  $\left|\frac{x}{2}\right| < 1 \Rightarrow |x| < 2$

(x)  $(1 + x - 2x^2)^{\frac{1}{2}}$

**Solution:**

$$\begin{aligned}
(1 + x - 2x^2)^{\frac{1}{2}} &= \left[1 + (x - 2x^2)\right]^{\frac{1}{2}} \\
&= 1 + \frac{1}{2}(x - 2x^2) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2} - 1\right)}{2!}(x - 2x^2)^2 + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} - 2\right)}{3!}(x - 2x^2)^3 + \dots \\
&= 1 + \frac{1}{2}(x - 2x^2) + \frac{\left(\frac{-1}{4}\right)}{2}(x - 2x^2)^2 + \frac{\frac{3}{8}}{6}(x - 2x^2)^3 + \dots \\
&= 1 + \frac{1}{2}(x - 2x^2) - \frac{1}{8}(x^2 - 4x^3 + 4x^4) + \frac{1}{16}(x^3 - 6x^4 + 12x^5 - 8x^6) + \dots \\
&= 1 + \frac{1}{2}x + \left(-1 - \frac{1}{8}\right)x^2 + \left(\frac{1}{2} + \frac{1}{16}\right)x^3 + \dots \\
&= 1 + \frac{1}{2}x - \frac{9}{8}x^2 + \frac{9}{16}x^3 + \dots
\end{aligned}$$

The expansion is valid if  $|x - 2x^2| < 1$

$$+(x - 2x^2) < 1$$

$$x - 2x^2 < 1$$

$$x - 2x^2 - 1 < 0$$

$$2x^2 - x + 1 > 0 \quad (i)$$

The inequality i) is not satisfied by any real value of  $x$

$$-(x - 2x^2) < 1$$

$$x - 2x^2 > -1$$

$$x - 2x^2 + 1 > 0$$

$$2x^2 - x - 1 < 0 \quad (ii)$$

$$2x^2 - 2x + x - 1 < 0$$

$$2x(x - 1) + 1(x - 1) < 0$$

$$(x - 1)(2x + 1) < 0$$

Case 1

$$x - 1 < 0, 2x + 1 > 0$$

$$x < 1, x > -\frac{1}{2}$$

Case 2

$$x - 1 > 0, 2x + 1 < 0$$

$$x > 1, x < -\frac{1}{2}$$

Which is not possible

Thus the expansion of  $(1+x-2x^2)^{-\frac{1}{2}}$  is valid if  $x \in \left(-\frac{1}{2}, 1\right)$  or  $-\frac{1}{2} < x < 1$

(xi)  $(1-2x+3x^2)^{\frac{1}{2}}$

**Solution:**

$$\begin{aligned}
 & (1-2x+3x^2)^{\frac{1}{2}} \\
 &= (1-(2x-3x^2))^{\frac{1}{2}} \\
 &= 1 + \left(\frac{-1}{3}\right)(-(2x-3x^2)) + \frac{\left(\frac{-1}{3}\right)\left(\frac{-1}{3}-1\right)(-(2x-3x^2))^2}{2!} + \frac{\left(\frac{-1}{3}\right)\left(\frac{-1}{3}-1\right)\left(\frac{-1}{3}-2\right)(-(2x-3x^2))^3}{3!} + \dots \\
 &= 1 + \frac{1}{3}(2x-3x^2) + \frac{4}{9} \times \frac{1}{2}(2x-3x^2)^2 + \frac{-28}{27} \times \frac{1}{6}(-(2x-3x^2)^3) + \dots \\
 &= 1 + \frac{1}{3}(2x-3x^2) + \frac{2}{9}(4x^2-12x^3+9x^4) + \frac{14}{81}(8x^3-36x^4+54x^5-27x^6) + \dots \\
 &= 1 + \frac{2}{3}x + \left(-1 + \frac{8}{9}\right)x^2 + \left(\frac{-8}{3}\right)x^3 + \frac{112}{81}x^4 + \dots \\
 &= 1 + \frac{2}{3}x - \frac{1}{9}x^2 - \frac{164}{81}x^3 + \dots
 \end{aligned}$$

The expansion is valid if  $|2x-3x^2| < 1$

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $  \begin{aligned}  & + (2x-3x^2) < 1 \\  & \quad 2x-3x^2 < 1 \\  & 3x^2-2x+1 > 0 \qquad (i)  \end{aligned}  $ | $  \begin{aligned}  & -(2x-3x^2) < 1 \\  & -2x+3x^2 < 1 \\  & 3x^2-2x-1 > 0 \qquad (ii) \\  & 3x^2-3x+x-1 < 0 \\  & 3x(x-1)+1(x-1) < 0 \\  & (x-1)(3x-1) < 0  \end{aligned}  $                                                                                                                                                                                                                                                                                                                                     |
| <p>The inequality i) is not satisfied by any real value of <math>x</math></p>                                  | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p style="text-align: center;">Case 1</p> <math display="block">x-1 &gt; 0, 3x+1 &lt; 0</math> <math display="block">x &gt; 1, x &lt; -\frac{1}{3}</math> <p style="text-align: center;">Which is not possible</p> </div> <div style="width: 45%;"> <p style="text-align: center;">Case 2</p> <math display="block">x-1 &lt; 0, 3x+1 &gt; 0</math> <math display="block">x &lt; 1, x &gt; -\frac{1}{3}</math> </div> </div> |

Thus the expansion of  $(1-2x+3x^2)^{\frac{1}{2}}$  is valid if  $x \in \left(-\frac{1}{3}, 1\right)$  or if  $-\frac{1}{3} < x < 1$

**Q.2 Using binominal theorem, find the value of the following to three places of decimals.**

(i)  $\sqrt{99}$

**Solution:**

$$\begin{aligned}
 \sqrt{99} &= (100-1)^{\frac{1}{2}} \\
 &= \left[100\left(1-\frac{1}{100}\right)\right]^{\frac{1}{2}}
 \end{aligned}$$

$$\begin{aligned}
&= (100)^{\frac{1}{2}} \left(1 - \frac{1}{100}\right)^{\frac{1}{2}} \\
&= 10 \left[ 1 + \frac{1}{2} \left(\frac{-1}{100}\right) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!} \left(\frac{-1}{100}\right)^2 + \dots \right] \\
&= 10 \left[ 1 - \frac{1}{2}(0.01) - \frac{1}{8}(0.0001) + \dots \right] \\
&= 10[1 - 0.005 - 0.0000125 + \dots] \\
&\approx 10[1 - 0.0050125] \\
&\approx 10[0.9949875] \\
&\approx 9.949875 \\
&\approx 9.950 \text{ convert to the three decimal}
\end{aligned}$$

(ii)  $(0.98)^{\frac{1}{2}}$

**Solution:**

$$\begin{aligned}
(0.98)^{\frac{1}{2}} &= (1 - 0.02)^{\frac{1}{2}} \\
&= 1 + \frac{1}{2}(-0.02) + \frac{\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)}{2!} \frac{1}{2}(-0.02)^2 + \frac{\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)}{3!}(-0.02)^3 + \dots \\
&= 1 - 0.1 - \frac{1}{8}(0.0004) - \frac{1}{16}(0.000008) + \dots \\
&\approx 1 - 0.1 - 0.00005 - 0.0000005 \\
&\approx 1 - 0.1000505 \\
&\approx 0.8999595 \\
&\approx 0.990
\end{aligned}$$

(iii)  $(1.03)^{\frac{1}{3}}$

**Solution:**

$$\begin{aligned}
(1.03)^{\frac{1}{3}} &= (1 + 0.03)^{\frac{1}{3}} \\
&= 1 + \frac{1}{3}(0.03) + \frac{\left(\frac{1}{3}\right)\left(\frac{1}{3}-1\right)}{2!}(0.03)^2 + \frac{\left(\frac{1}{3}\right)\left(\frac{1}{3}-1\right)\left(\frac{1}{3}-2\right)}{3!}(0.03)^3 + \dots \\
&= 1 + 0.01 + \frac{\left(\frac{1}{3}\right)\left(\frac{-2}{3}\right)}{2} \frac{1}{2}(0.0009) + \frac{\left(\frac{1}{3}\right)\left(\frac{-2}{3}\right)\left(\frac{-5}{3}\right)}{6} \frac{1}{6}(0.000027) + \dots \\
&= 1 + 0.01 - \frac{1}{9}(0.0009) + \frac{5}{81}(0.000027) + \dots \\
&\approx 1 + 0.1 - 0.0001 + 0.0001 \\
&\approx 1.010
\end{aligned}$$

(iv)  $\sqrt[3]{65}$

**Solution:**

$$\begin{aligned}
\sqrt[3]{65} &= (64+1)^{\frac{1}{3}} \\
&= \left[ 64 \left( 1 + \frac{1}{64} \right) \right]^{\frac{1}{3}} \\
&= (64)^{\frac{1}{3}} \left( 1 + \frac{1}{64} \right)^{\frac{1}{3}} \\
&= 4 \left( 1 + \frac{1}{64} \right)^{\frac{1}{3}} \\
&= 4 \left[ 1 + \frac{1}{3} \left( \frac{1}{64} \right) + \frac{\left( \frac{1}{3} \right) \left( \frac{1}{3} - 1 \right)}{2!} \left( \frac{1}{64} \right)^2 + \dots \right] \\
&= 4 \left[ 1 + \frac{1}{3} (0.015625) - \frac{1}{9} (0.015625)^2 + \dots \right] \\
&\approx 4[1 + 0.005208 - 0.000027] \\
&\approx 4(1.005181) \\
&\approx 4.020724 \\
&\approx 4.021
\end{aligned}$$

**(v)**  $\sqrt[4]{17}$ **Solution:**

$$\begin{aligned}
\sqrt[4]{17} &= (17)^{\frac{1}{4}} \\
&= (16+1)^{\frac{1}{4}} \\
&= \left[ 16 \left( 1 + \frac{1}{16} \right) \right]^{\frac{1}{4}} \\
&= (16)^{\frac{1}{4}} \left( 1 + \frac{1}{16} \right)^{\frac{1}{4}} \\
&= 2 \left( 1 + \frac{1}{16} \right)^{\frac{1}{4}} \\
&= 2 \left[ 1 + \frac{1}{4} \left( \frac{1}{16} \right) + \frac{\left( \frac{1}{4} \right) \left( \frac{1}{4} - 1 \right)}{2!} \left( \frac{1}{16} \right)^2 + \dots \right]
\end{aligned}$$

$$= 2 \left[ 1 + \frac{1}{64} + \left( \frac{1}{4} \right) \left( \frac{-3}{4} \right) \frac{1}{2} \left( \frac{1}{16} \right)^2 + \dots \right]$$

$$= 2 \left[ 1 + \frac{1}{64} - \frac{3}{2} \times \left( \frac{1}{64} \right)^2 + \dots \right]$$

$$\approx 2[1 + 0.015625 - 0.000366]$$

$$\approx 2[1.015259]$$

$$\approx 2.030518$$

$$\approx 2.031$$

(vi)  ${}^5\sqrt{31}$

**Solution:**

$${}^5\sqrt{31} = (31)^{\frac{1}{5}}$$

$$= (32 - 1)^{\frac{1}{5}}$$

$$= \left[ 32 \left( 1 - \frac{1}{32} \right) \right]^{\frac{1}{5}}$$

$$= (32)^{\frac{1}{5}} \left( 1 - \frac{1}{32} \right)^{\frac{1}{5}}$$

$$= 2 \left( 1 - \frac{1}{32} \right)^{\frac{1}{5}}$$

$$= 2 \left[ 1 + \frac{1}{5} \left( \frac{-1}{32} \right) + \frac{\left( \frac{1}{5} \right) \left( \frac{1}{5} - 1 \right)}{2!} \left( \frac{-1}{32} \right)^2 + \dots \right]$$

$$= 2 \left[ 1 - \frac{1}{5 \times 32} + \frac{1}{5} \times \frac{-4}{5} \times \frac{1}{2} \times \left( \frac{1}{32} \right)^2 + \dots \right]$$

$$= 2 \left[ 1 - \frac{1}{10} \times \frac{1}{16} - 2 \left( \frac{1}{10} \times \frac{1}{16} \right)^2 + \dots \right]$$

$$= 2[1 - 0.00625 - 2(0.0000390625) + \dots]$$

$$\approx 2(1 - 0.00625 - 0.000078)$$

$$\approx 2(0.993672)$$

$$\approx 1.987344$$

$$\approx 1.987$$

(vii)  $\frac{1}{\sqrt[3]{998}}$

**Solution:**



$$\begin{aligned}
\frac{1}{\sqrt[3]{998}} &= (998)^{-\frac{1}{3}} \\
&= (1000 - 2)^{-\frac{1}{3}} \\
&= \left[ (1000) \left( 1 - \frac{2}{1000} \right) \right]^{-\frac{1}{3}} \\
&= (1000)^{-\frac{1}{3}} \left( 1 - \frac{1}{500} \right)^{-\frac{1}{3}} \\
&= (10)^{-1} \left[ 1 + \left( \frac{-1}{3} \right) \left( \frac{-1}{500} \right) + \frac{\left( \frac{1}{3} \right) \left( \frac{-1}{3} - 1 \right)}{2!} \left( -\frac{1}{500} \right)^2 + \dots \right] \\
&\approx \frac{1}{10} [1 + 0.0006667 + 0.00000080] \\
&\approx \frac{1}{10} (1.0006675) \\
&\approx 0.10006675 \\
&\approx 0.100
\end{aligned}$$

(viii)  $\frac{1}{\sqrt[5]{252}}$

**Solution:**

$$\begin{aligned}
\frac{1}{\sqrt[5]{252}} &= (252)^{-\frac{1}{5}} \\
&= (243 + 9)^{-\frac{1}{5}} \\
&= \left[ 243 \left( 1 + \frac{9}{243} \right) \right]^{-\frac{1}{5}} \\
&= (243)^{-\frac{1}{5}} \left( 1 + \frac{1}{27} \right)^{-\frac{1}{5}} \\
&= \frac{1}{3} \left[ 1 + \left( \frac{-1}{5} \right) \left( \frac{1}{27} \right) + \frac{\left( \frac{-1}{5} \right) \left( \frac{-1}{5} - 1 \right)}{2!} \left( \frac{1}{27} \right)^2 + \dots \right] \\
&= \frac{1}{3} \left[ 1 - \frac{1}{5 \times 27} + \left( \frac{-1}{5} \right) \left( \frac{-6}{5} \right) \frac{1}{2} \left( \frac{1}{27} \right)^2 + \dots \right] \\
&= \frac{1}{3} \left[ 1 - \frac{1}{5 \times 27} + 3 \left( \frac{1}{5 \times 27} \right)^2 + \dots \right]
\end{aligned}$$

$$\approx \frac{1}{3} [1 - 0.0074074 + 3(0.00005487)]$$

$$\approx \frac{1}{3} (0.992757)$$

$$\approx 0.330919$$

$$\approx 0.331$$

(ix)  $\frac{\sqrt{7}}{\sqrt{8}}$

**Solution:**

$$\begin{aligned} \sqrt{\frac{7}{8}} &= \left(1 - \frac{1}{8}\right)^{\frac{1}{2}} \\ &= 1 + \frac{1}{2} \left(\frac{-1}{8}\right) + \frac{\frac{1}{2} \left(\frac{1}{2} - 1\right)}{2!} \left(\frac{-1}{8}\right)^2 + \dots \\ &= 1 - \frac{1}{16} + \left(\frac{1}{2}\right) \left(\frac{-1}{2}\right) + \frac{1}{2} \times \frac{1}{64} + \dots \\ &= 1 - \frac{1}{16} - \frac{1}{8 \times 64} + \dots \\ &\approx 1 - 0.0625 - 0.0193 \\ &\approx 0.935448 \\ &\approx 0.935 \end{aligned}$$

(x)  $(0.998)^{\frac{-1}{3}}$

**Solution:**

$$\begin{aligned} (0.998)^{\frac{-1}{3}} &= (1 - 0.002)^{\frac{-1}{3}} \\ &= 1 + \left(-\frac{1}{3}\right) (-0.002) + \frac{\left(-\frac{1}{3}\right) \left(\frac{-1}{3} - 1\right)}{2!} (-0.002)^2 + \dots \\ &= 1 + \frac{1}{3} (0.002) + \frac{2}{9} (0.000004) + \dots \\ &\approx 1 + 0.000666 \\ &\approx 1.001 \end{aligned}$$

(xi)  $\frac{1}{\sqrt[6]{486}}$

**Solution:**

$$\begin{aligned} \frac{1}{\sqrt[6]{486}} &= (486)^{\frac{-1}{6}} = (729 - 243)^{\frac{-1}{6}} \\ &= \left[729 \left(1 - \frac{243}{729}\right)\right]^{\frac{-1}{6}} \end{aligned}$$

$$\begin{aligned}
&= (3^6)^{-\frac{1}{6}} \left[ 1 + \left( \frac{-1}{6} \right) \left( -\frac{1}{3} \right) + \frac{\left( \frac{-1}{6} \right) \left( \frac{-7}{6} \right)}{2!} \left( \frac{1}{3} \right)^2 + \dots \right] \\
&= 3^{-1} \left[ 1 + \frac{1}{18} + \frac{7}{36} \cdot \frac{1}{2} \cdot \frac{1}{9} + \dots \right] \\
&= \frac{1}{3} \left[ 1 + 0.05556 + \frac{7}{2} (0.003086) + \dots \right] \\
&\approx \frac{1}{3} [1 + 0.05556 + 0.0108] \\
&\approx \frac{1}{3} [1.06896] \\
&\approx 0.35632 \\
&\approx 0.3563
\end{aligned}$$

(xii)  $(1280)^{\frac{1}{4}}$

**Solution:**

$$\begin{aligned}
(1280)^{\frac{1}{4}} &= (1296 - 16)^{\frac{1}{4}} \\
&= \left[ 1296 \left( 1 - \frac{16}{1296} \right) \right]^{\frac{1}{4}} \\
&= (1296)^{\frac{1}{4}} \left[ 1 - \frac{1}{81} \right]^{\frac{1}{4}} \\
&= 6 \left[ 1 + \frac{1}{4} \left( \frac{-1}{81} \right) + \frac{\left( \frac{1}{4} \right) \left( \frac{1}{4} - 1 \right)}{2!} \left( -\frac{1}{81} \right)^2 + \dots \right] \\
&\approx 6 \left[ 1 - 0.003086 - \frac{3}{2} (0.0000095) \right] \\
&\approx 6 [1 - 0.0031] \\
&\approx 6 (0.9969) \\
&\approx 5.981 \\
&\approx 5.981
\end{aligned}$$

**Q.3 Find the coefficient of  $x^n$  in the expansion of**

(i)  $\frac{1+x^2}{(1+x)^2}$

**Solution:**

$$\frac{1+x^2}{(1+x)^2} = (1+x^2)(1+x)^{-2}$$

From  $(1+x)^2$  firstly we find the coefficient of  $x^{n-2}$  and  $x^n$

As we know that  $(r+1)^{th}$  term of  $(1+x)^n$  is

$$T_{r+1} = \frac{n(n-1)(n-2)(n-3)\dots(n-(r-1))x^r}{r!}$$

For  $x^n$  put  $n = -2, r = n$  we get

$$\begin{aligned} T_{n+1} &= \frac{-2(-2-1)(-2-2)(-2-3)\dots(-2-(n-1))x^n}{n!} \\ &= \frac{(-2)(-3)(-4)\dots(-n-1)x^n}{n!} \\ &= \frac{(-1)^n(2)(3)(4)\dots(n+1)x^n}{n!} \\ &= (-1)^n \frac{(n+1)!}{n!} x^n \end{aligned}$$

$$T_{n+1} = (-1)^n (n+1)x^n$$

So in  $(1+x)^{-2}$  coefficient of  $x^n$  is  $(-1)^n (n+1)$

So coefficient of  $x^{n-2}$  is  $(-1)^{n-2} (n-1)x^{n-2}$

$$\begin{aligned} &= (-1)^n (n+1)x^n + (-1)^{n-2} (n-1)x^2 \times x^{n-2} \\ &= (-1)^n \{(n+1) + (-1)^{-2} (n-1)\} x^n \\ &= (-1)^n (2n)x^n \end{aligned}$$

Hence coefficient of  $x^n$  is  $(-1)^n (2n)$

$$(ii) \quad \frac{(1+x)^2}{(1-x)^2}$$

**Solution:**

$$\begin{aligned} \frac{(1+x)^2}{(1-x)^2} &= (1+x)^2 (1-x)^{-2} \\ &= (1+2x+x^2)(1-x)^{-2} \end{aligned}$$

From  $(1-x)^2$  firstly we find the coefficient of  $x^{n-2}, x^{n-1}$  and  $x^n$

As we know that  $(r+1)^{th}$  term of  $(1+x)^n$  is

$$T_{r+1} = \frac{n(n-1)(n-2)(n-3)\dots(n-(r-1))x^r}{r!}$$

For  $x^n$  put  $n = -2, r = n, x = -x$  we get

$$T_{n+1} = \frac{(-2)(-2-1)(-2-2)(-2-3)\dots(-2-(n-1))(-x)^n}{n!}$$

$$T_{n+1} = \frac{(-2)(-3)(-4)(-5)\dots(-1-n)(-x)^n}{n!}$$

There are  $n$  factors in the numerators, so taking  $-1$  from each factors we get;

$$T_{n+1} = \frac{(-1)^n (2 \times 3 \times 4 \times 5 \times \dots \times (n+1)) x^n (-1)^n}{n!}$$

$$T_{n+1} = (-1)^{n+n} \frac{(n+1)!}{n!} x^n$$

$$T_{n+1} = (-1)^{2n} \times \frac{(n+1)n!}{n!} x^n$$

So in  $(1-x)^{-2}$  coefficient of  $x^n$  is  $(n+1)$ .

So coefficient of  $x^{n-1}$  is  $n$

Coefficient of  $x^{n-2}$  is  $(n-1)$

Now in  $(1+2x+x^2)(1-x)^{-2}$  term involving  $x^n$  is

$$= (n+1)x^n + 2 \times n \times x^{n-1} \times x + 1 \times (n-1)x^{n-2} \times x^2$$

$$= x^n \{n+1+2n+n-1\}$$

$$= \{4n\} x^n \text{ so coefficient of } x^n \text{ is } 1+n$$

(iii)  $\frac{(1+x)^3}{(1-x)^2}$

**Solution:**

$$\frac{(1+x)^3}{(1-x)^2} = (1+x)^3 (1-x)^2$$

$$= (1+3x^2+3x+x^3)(1-x)^2 \text{ -----(i)}$$

In order to find coefficient of  $x^n$  in  $\frac{(1+x)^3}{(1-x)^2}$

We have to need coefficient of  $x^n, x^{n-1}, x^{n-2}, x^{n-3}$  from  $(1-x)^2$

As we know that  $(r+1)^{th}$  term of  $(1+x)^n$  is

$$T_{r+1} = \frac{n(n-1)(n-2)(n-3)\dots(n-(r-1))x^r}{r!}$$

So put  $n = -2, x = -x, r = n$  we get

$$T_{n+1} = \frac{(-2)(-2-1)(-2-2)(-2-3)\dots(-2-(n-1))(-x)^n}{n!}$$

$$\begin{aligned}
 T_{n+1} &= \frac{(-2)(-3)(-4)(-5)....(-1-n)x^n \times (-1)^n}{n!} \\
 &= \frac{(-1)^n (2 \times 3 \times 4 \times ....(n+1))x^n \times (-1)^n}{n!} \\
 &= \left((-1)^n\right)^2 \frac{(n+1)!}{n!} x^n
 \end{aligned}$$

$$T_{n+1} = (n+1)x^n$$

So coefficient of  $x^{n-1}$  is  $n$

Coefficient of  $x^{n-2}$  is  $n-1$

Coefficient of  $x^{n-3}$  is  $n-2$

Now the term Involving  $x^n$  in  $\frac{(1+x)^3}{(1-x)^2}$  is

$$\begin{aligned}
 &= 1 \times (n+1)x^n + 3x^2 \times (n-1)x^{n-2} + 3x \times (n)x^{n-1} + (n-2)x^{n-3} \times x^3 \\
 &= (n+1)x^n + (3n-3)x^n + 3nx^n + (n-2)x \\
 &= (8n-4)x^n \\
 &= 4(2n-1)x^n
 \end{aligned}$$

Hence coefficient of  $x^n$  is  $4(2n-1)$

(iv)  $\frac{(1+x)^2}{(1-x)^3}$

**Solution:**

$$\begin{aligned}
 \frac{(1+x)^2}{(1-x)^3} &= (1+x)^2 (1-x)^{-3} \\
 &= (1+2x+x^2)(1-x)^{-3}
 \end{aligned}$$

From  $(1-x)^{-3}$  firstly we find the coefficient of  $x^{n-2}$ ,  $x^{n-1}$  and  $x^n$

As we know that  $(r+1)^{th}$  term of  $(1+x)^n$  is

$$T_{r+1} = \frac{n(n-1)(n-2)(n-3)....(n-(r-1))x^r}{r!}$$

For  $x^n$  put  $n=-3$ ,  $r=n$  and  $x=-x$  we get

$$\begin{aligned}
 T_{n+1} &= \frac{-3(-3-1)(-3-2)(-3-3)....(-3-(n-1))(-x)^n}{n!} \\
 T_{n+1} &= \frac{(-3)(-4)(-5)....(-3-n+1)(-x)^n}{n!}
 \end{aligned}$$

$$T_{n+1} = \frac{(-1)^n (2)(3)(4)(5) \dots (n+2) x^n (-1)^n}{2n!}$$

$$T_{n+1} = \frac{(-1)^{2n} (n+2)! x^n}{2n!}$$

$$T_{n+1} = \frac{(n+2)(n+1)n! x^n}{2n!}$$

$$T_{n+1} = (n+2)(n+1)x^n$$

So coefficient of  $x^n$  is  $\frac{(n+2)(n+1)}{2}$

Coefficient of  $x^{n-1}$  is  $\frac{(n+1)n}{2}$  and coefficient of  $x^{n-2}$  is  $\frac{n(n-1)}{2}$

Now the term Involving  $x^n$  in  $(1+2x+x^2)(1-x)^{-3}$  is

$$= \frac{(n+2)(n+1)}{2} x^n + \frac{2(n+1)n}{2} x \cdot x^{n-1} + \frac{n(n-1)}{2} x^2 \cdot x^{n-2}$$

$$= \left\{ \frac{(n+2)(n+1)}{2} + \frac{2(n+1)n}{2} + \frac{n(n-1)}{2} \right\} x^n$$

$$= \frac{1}{2} \{ n^2 + 3n + 2 + 2n^2 + 2n + n^2 - n \} x^n$$

$$= \frac{1}{2} \{ 4n^2 + 4n + 2 \} x^n$$

$$= (2n^2 + 2n + 1) x^n$$

Thus coefficient of  $x^n$  is  $2n^2 + 2n + 1$

(v)  $(1 - x + x^2 - x^3 + \dots)^2$

**Solution:**

As we know that

$$1 - x + x^2 - x^3 + \dots = (1+x)^{-1}$$

$$\Rightarrow (1 - x + x^2 - x^3 + \dots)^2 = \left( (1+x)^{-1} \right)^2 = (1+x)^{-2}$$

Now we find the coefficient of  $x^n$  in  $(1+x)^{-2}$  by using formula

$$T_{r+1} = \frac{n(n-1)(n-2)(n-3) \dots (n-(r-1)) x^r}{r!}$$

Put  $n = -2$ ,  $r = n$  we get

$$T_{n+1} = \frac{(-2)(-2-1)(-2-2)(-2-3) \dots (-2-(n-1)) x^n}{n!}$$

$$= \frac{(-2)(-3)(-4)(-5) \dots (-1-n) x^n}{n!}$$

$$\begin{aligned}
 &= \frac{(-1)^n [2 \times 3 \times 4 \times 5 \times \dots \times (n+1)] x^n}{n!} \\
 &= \frac{(-1)^n (n+1)!}{n!} x^n \\
 &= (-1)^n (n+1) x^n
 \end{aligned}$$

Thus coefficient of  $x^n$  is  $(-1)^n (n+1)$ .

**Q.4** If  $x$  is so small that its square and higher powers can be neglected then show that

(i)  $\frac{1-x}{\sqrt{1+x}} \approx 1 - \frac{3}{2}x$

**Solution:**

$$\begin{aligned}
 \text{L.H.S} &= \frac{1-x}{\sqrt{1+x}} = (1-x)(1+x)^{-\frac{1}{2}} \\
 &= (1-x) \left\{ 1 + \left( \frac{-1}{2} \right) x \right\} \text{ by neglecting } x^2 \text{ and highest power of } x. \\
 &\approx 1 - \frac{1}{2}x - x \text{ by neglecting } x^2 \\
 &\approx 1 - \frac{3}{2}x \\
 &\approx \text{R.H.S}
 \end{aligned}$$

(ii)  $\frac{\sqrt{1+2x}}{1-x} \approx 1 + \frac{3}{2}x$

**Solution:**

$$\begin{aligned}
 \text{L.H.S} &= \frac{\sqrt{1+2x}}{1-x} \\
 &= (1+2x)^{\frac{1}{2}} (1-x)^{-\frac{1}{2}} \\
 &= \left( 1 + \frac{1}{2}(2x) \right) \left( 1 - \left( \frac{-1}{2} \right) x \right) \text{ by neglecting } x^2 \text{ and higher powers of } x. \\
 &\approx (1+x) \left( 1 + \frac{x}{2} \right) \\
 &\approx 1 + \frac{x}{2} + x \text{ by neglecting } x^2 \\
 &\approx 1 + \frac{3}{2}x \\
 &\approx \text{R.H.S}
 \end{aligned}$$

(iii)  $\frac{(9+7x)^{\frac{1}{2}} - (16+3x)^{\frac{1}{2}}}{4+5x} \approx \frac{1}{4} - \frac{17}{384}x$

**Solution:**



$$\begin{aligned}
 \text{L.H.S} &= \frac{(9+7x)^{\frac{1}{2}} - (16+3x)^{\frac{1}{4}}}{(4+5x)} \\
 &= \left\{ 9^{\frac{1}{2}} \left( 1 + \frac{7}{9}x \right)^{\frac{1}{2}} - (16)^{\frac{1}{4}} \left( 1 + \frac{3x}{16} \right)^{\frac{1}{4}} \right\} (4+5x)^{-1} \\
 &= \left\{ 3 \left( 1 + \frac{7}{9}x \right)^{\frac{1}{2}} - 2 \left( 1 + \frac{3x}{16} \right)^{\frac{1}{4}} \right\} 4^{-1} \left\{ 1 + \frac{5}{4}x \right\}^{-1} \\
 &= \left\{ 3 \left( 1 + \frac{7}{9}x \cdot \frac{1}{2} \right) - 2 \left( 1 + \frac{1}{4} \left( \frac{3x}{16} \right) \right) \right\} \frac{\left( 1 - \frac{5}{4}x \right)}{4}
 \end{aligned}$$

By neglecting  $x^2$  and higher powers of  $x$ .

$$\begin{aligned}
 &\approx \left\{ 3 + \frac{7x}{6} - 2 - \frac{3x}{32} \right\} \frac{\left( 1 - \frac{5}{4}x \right)}{4} \\
 &\approx \frac{\left\{ 1 + \frac{103}{96}x \right\} \left\{ 1 - \frac{5}{4}x \right\}}{4} \\
 &\approx \frac{1 - \frac{5}{4}x + \frac{103}{96}x}{4} \quad \text{by neglecting } x^2 \\
 &\approx \frac{1 - \frac{17}{96}x}{4} \\
 &\approx \frac{1}{4} - \frac{17}{384}x \\
 &\approx \text{R.H.S}
 \end{aligned}$$

$$(iv) \quad \frac{\sqrt{4+x}}{(1-x)^3} \approx 2 + \frac{25}{4}x$$

**Solution:**

$$\begin{aligned} \text{L.H.S} &= \frac{\sqrt{4+x}}{(1-x)^3} \\ &= (4+x)^{\frac{1}{2}}(1-x)^{-3} \\ &= 4^{\frac{1}{2}} \left(1 + \frac{x}{4}\right)^{\frac{1}{2}} (1-x)^{-3} \\ &= 2 \left(1 + \frac{1}{2} \left(\frac{x}{4}\right)\right) (1+3x) \text{ by neglecting } x^2 \text{ and highest power of } x \\ &\approx 2 \left(1 + \frac{x}{8}\right) (1+3x) \\ &\approx 2 \left(1 + 3x + \frac{x}{8}\right) \text{ by neglecting } x^2 \\ &\approx 2 + \frac{25}{4}x \\ &\approx \text{R.H.S} \end{aligned}$$

$$(v) \quad \frac{(1+x)^{\frac{1}{2}}(4-3x)^{\frac{3}{2}}}{(8+5x)^{\frac{1}{3}}} \approx 4 \left(1 - \frac{5x}{6}\right)$$

**Solution:**

$$\begin{aligned} \text{L.H.S} &= \frac{(1+x)^{\frac{1}{2}}(4-3x)^{\frac{3}{2}}}{(8+5x)^{\frac{1}{3}}} \\ &= \left\{ (1+x)^{\frac{1}{2}} (4-3x)^{\frac{3}{2}} \right\} (8+5x)^{-\frac{1}{3}} \\ &= \left\{ (1+x)^{\frac{1}{2}} \cdot (4)^{\frac{3}{2}} \left(1 - \frac{3}{4}x\right)^{\frac{3}{2}} \right\} \times (8)^{-\frac{1}{3}} \left(1 + \frac{5}{8}x\right)^{-\frac{1}{3}} \\ &= \left\{ \left(1 + \frac{1}{2}x\right) \cdot 2^3 \times \left(1 - \frac{3}{2} \left(\frac{3}{4}x\right)\right) \right\} \times (2^3)^{-\frac{1}{3}} \left(1 - \frac{1}{3} \left(\frac{5}{8}x\right)\right) \text{ by neglecting } x^2, x^3, \dots \\ &\approx 8 \left\{ \left(1 + \frac{1}{2}x\right) \left(1 - \frac{9}{8}x\right) \right\} \times 2^{-1} \left(1 - \frac{5}{24}x\right) \\ &\approx \frac{8}{2} \left\{ 1 - \frac{9}{8}x + \frac{1}{2}x \right\} \left(1 - \frac{5}{24}x\right) \text{ by neglecting } x^2 \\ &\approx 4 \left\{ 1 - \frac{5}{8}x \right\} \left(1 - \frac{5}{24}x\right) \end{aligned}$$

$$\approx 4 \left( 1 - \frac{5}{24}x - \frac{5}{8}x \right) \text{ by neglecting } x^2$$

$$\approx 4 \left( 1 - \frac{5}{6}x \right)$$

$\approx \text{R.H.S}$

$$(vi) \quad \frac{(1-x)^{\frac{1}{2}}(9-4x)^{\frac{1}{2}}}{(8+3x)^{\frac{1}{3}}} \approx \frac{3}{2} - \frac{61}{48}x$$

**Solution:**

$$\begin{aligned} \text{L.H.S} &= \frac{(1-x)^{\frac{1}{2}}(9-4x)^{\frac{1}{2}}}{(8+3x)^{\frac{1}{3}}} \\ &= \left\{ (1-x)^{\frac{1}{2}} \cdot 9^{\frac{1}{2}} \left( 1 - \frac{4}{9}x \right)^{\frac{1}{2}} \right\} (8+3x)^{-\frac{1}{3}} \\ &= 3 \left( 1 - \frac{1}{2}x + \dots \right) \left( 1 - \frac{1}{2} \left( \frac{4}{9}x \right) + \dots \right) \times 8^{-\frac{1}{3}} \left( 1 + \frac{3x}{8} \right)^{-\frac{1}{3}} \text{ by neglecting } x^2, x^3, \dots \\ &\approx 3 \left( 1 - \frac{1}{2}x \right) \left( 1 - \frac{2}{9}x \right) \times 2^{-1} \left( 1 - \frac{1}{3} \times \frac{3x}{8} + \dots \right) \\ &\approx \frac{3}{2} \left( 1 - \frac{2}{9}x - \frac{1}{2}x \right) \left( 1 - \frac{x}{8} \right) \text{ by neglecting } x^2. \\ &\approx \frac{3}{2} \left( 1 - \frac{13}{18}x \right) \left( 1 - \frac{x}{8} \right) \\ &\approx \frac{3}{2} \left( 1 - \frac{x}{8} - \frac{13}{18}x \right) \text{ by neglecting } x^2 \\ &\approx \frac{3}{2} \left( 1 - \frac{61}{72}x \right) \\ &\approx \frac{3}{2} - \frac{61}{48}x \\ &\approx \text{R.H.S} \end{aligned}$$

$$(vii) \quad \frac{\sqrt{4-x} + (8-x)^{\frac{1}{3}}}{(8-x)^{\frac{1}{3}}} \approx 2 - \frac{1}{2}x$$

**Solution:**

$$\text{L.H.S} = \frac{(4-x)^{\frac{1}{2}} + (8-x)^{\frac{1}{3}}}{(8-x)^{\frac{1}{3}}}$$

$$\begin{aligned}
&= \frac{(4-x)^{\frac{1}{2}}}{(8-x)^{\frac{1}{3}}} + \frac{(8-x)^{\frac{1}{3}}}{(8-x)^{\frac{1}{3}}} \\
&= (4-x)^{\frac{1}{2}} (8-x)^{-\frac{1}{3}} + 1 \\
&= 4^{\frac{1}{2}} \left(1 - \frac{x}{4}\right)^{\frac{1}{2}} \times 8^{-\frac{1}{3}} \left(1 - \frac{x}{8}\right)^{-\frac{1}{3}} + 1 \\
&= 2 \left(1 - \frac{1}{2} \left(\frac{x}{4}\right) + \dots\right) \times 2^{-1} \left(1 + \frac{1}{3} \left(\frac{x}{8}\right) + \dots\right) + 1 \text{ by neglecting } x^2, x^3, \dots \\
&\approx \frac{\left(2 - \frac{x}{4}\right) \left(1 + \frac{x}{24}\right)}{2} + 1 \\
&\approx \frac{2 + \frac{x}{12} - \frac{x}{4}}{2} + 1 \quad \text{By neglecting } x^2 \\
&\approx \frac{2 - \frac{1}{6}x}{2} + 1 \\
&\approx 1 - \frac{1}{12}x + 1 \\
&\approx 2 - \frac{1}{12}x \\
&\approx \text{R.H.S}
\end{aligned}$$

**Q.5** If  $x$  is so small that its cube and higher power can be neglected, then show that

$$(i) \quad \sqrt{1-x-2x^2} \approx 1 - \frac{1}{2}x - \frac{9}{8}x^2$$

**Solution:**

$$\begin{aligned} \text{L.H.S} &= \sqrt{1-x-2x^2} \\ &= \left(1 - (x+2x^2)\right)^{\frac{1}{2}} \\ &= 1 - \frac{1}{2}(x+2x^2) + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)(x+2x^2)^2}{2!} \text{ by neglecting } x^3, x^4, \dots \\ &= 1 - \frac{1}{2}x - x^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)(x^2)}{2} \text{ by neglecting } x^3, x^4 \dots \\ &\approx 1 - \frac{1}{2}x - x^2 - \frac{1}{8}x^2 \\ &\approx 1 - \frac{1}{2}x - \frac{9}{8}x^2 \\ &\approx \text{R.H.S} \end{aligned}$$

$$(ii) \quad \sqrt{\frac{1+x}{1-x}} \approx 1+x+\frac{1}{2}x^2$$

**Solution:**

$$\begin{aligned} \text{L.H.S} &= \sqrt{\frac{(1+x)}{(1-x)} \times \frac{1+x}{1+x}} \\ &= (1+x)(1-x^2)^{-\frac{1}{2}} \\ &= (1+x)\left(1+\frac{1}{2}x^2\right) \text{ by neglecting } x^3, x^4 \dots \\ &\approx (1+x)\left(1+\frac{1}{2}x^2\right) \\ &\approx 1+x+\frac{1}{2}x^2 \text{ by neglecting } x^3 \\ &\approx 1+x+\frac{1}{2}x^2 \\ &\approx \text{R.H.S} \end{aligned}$$

**Q.6** If  $x$  is nearly equal to 1, then prove that  $px^p - qx^q \approx (p-q)x^{p+q}$

**Proof:** As  $x$  is nearly equal to 1, so

Let  $x = 1+h$  where  $h$  is so small such that  $h^2, h^3, \dots$  are neglected.

$$\begin{aligned}
 \text{L.H.S} &= px^p - qx^q \\
 &= p(1+h)^p - q(1+h)^q \\
 &= p(1+ph) - q(1+qh) \text{ by neglecting } h^2, h^3 \dots \\
 &\approx p + p^2h - q - q^2h \\
 &\approx (p-q) + (p^2 - q^2)h \\
 &\approx (p-q) + (p-q)(p+q)h \\
 &\approx (p-q)[1 + (p+q)h] \\
 &\approx (p-q)(1+h)^{p+q} \\
 &\approx (p-q)(x)^{p+q} \\
 &\approx \text{R.H.S}
 \end{aligned}$$

**Q.7** If  $p-q$  is small when compared with  $p$  or  $q$  show that

$$\frac{(2n+1)p + (2n-1)q}{(2n-1)p + (2n+1)q} \approx \left( \frac{p+q}{2q} \right)^{\frac{1}{n}}$$

**Proof:** Let  $p-q = h \Rightarrow p = q+h$  where  $h$  is very small such that  $h^2, h^3, \dots$  are neglected

$$\begin{aligned}
 \text{L.H.S} &= \frac{(2n+1)p + (2n-1)q}{(2n-1)p + (2n+1)q} \\
 &= \frac{(2n+1)(q+h) + (2n-1)q}{(2n-1)(q+h) + (2n+1)q} \\
 &= \frac{2nq + 2nh + q + h + 2nq - q}{2nq + 2nh - q - h + 2nq + q} \\
 &= \frac{4nq + 2nh + h}{4nq + 2nh - h} \\
 &= \frac{4nq + h(2n+1)}{4nq + h(2n-1)} \\
 &= \frac{4nq \left\{ 1 + \left( \frac{2n+1}{4nq} \right) h \right\}}{4nq \left\{ 1 + \left( \frac{2n-1}{4nq} \right) h \right\}} \\
 &= \left\{ 1 + \left( \frac{2n+1}{4nq} \right) h \right\} \left\{ 1 + \left( \frac{2n-1}{4nq} \right) h \right\}^{-1}
 \end{aligned}$$

$$\begin{aligned}
&= \left\{ 1 + \left( \frac{2n+1}{4nq} \right) h \right\} \left\{ 1 - \left( \frac{2n-1}{4nq} \right) h \right\} \text{ by neglecting } h^2, h^3, \dots \\
&\approx 1 - \left( \frac{2n-1}{4nq} \right) h + \left( \frac{2n+1}{4nq} \right) h \text{ by neglecting } h^2 \\
&\approx 1 - h \left( \frac{(2n-1)}{4nq} - \frac{(2n+1)}{4nq} \right) \\
&\approx 1 - \frac{h}{4nq} (2n-1-2n-1) \\
&\approx 1 - \frac{h}{4nq} (-2) \\
&\approx 1 + \frac{h}{2nq} \\
\text{R.H.S} &= \left( \frac{p+q}{2q} \right)^{\frac{1}{n}} \\
&\approx \left( \frac{q+h+q}{2q} \right)^{\frac{1}{n}} \quad \because p = q + h \\
&\approx \left( \frac{2q+h}{2q} \right)^{\frac{1}{n}} \\
&\approx \left( 1 + \frac{h}{2q} \right)^{\frac{1}{n}} \\
&\approx 1 + \frac{h}{2nq} \text{ by neglecting } h^2, h^3, \dots
\end{aligned}$$

Hence L.H.S  $\approx$  R.H.S

**Q.8 Show that**

$$\left[ \frac{n}{2(n+N)} \right]^{\frac{1}{2}} \approx \frac{8n}{9n-N} - \frac{n+N}{4n} \text{ where } n \text{ and } N \text{ are nearly equal.}$$

**Proof:** Here  $N - n = h \Rightarrow N = n + h$  where  $h$  is so small, such that  $h^2, h^3, \dots$  are neglected

$$\begin{aligned}
\text{L.H.S} &= \left[ \frac{n}{2(n+N)} \right]^{\frac{1}{2}} \\
&= \left[ \frac{n}{2(n+n+h)} \right]^{\frac{1}{2}}
\end{aligned}$$

$$\begin{aligned}
&= \left[ \frac{n}{2(2n+h)} \right]^{\frac{1}{2}} \\
&= \left[ \frac{n}{2 \times 2n \left( 1 + \frac{h}{2n} \right)} \right]^{\frac{1}{2}} \\
&= \left[ \frac{1}{4 \left( 1 + \frac{h}{2n} \right)} \right]^{\frac{1}{2}} \\
&= \frac{1}{2} \left( 1 + \frac{h}{2n} \right)^{-\frac{1}{2}} \\
&= \frac{1}{2} \left( 1 - \frac{1}{2} \left( \frac{h}{2n} \right) \right) \text{ by neglecting } h^2, h^3, \dots \\
&\approx \frac{1}{2} - \frac{1}{8} \frac{h}{n}
\end{aligned}$$

$$\text{R.H.S} = \frac{8n}{9n-N} - \frac{n+N}{4n}$$

$$\text{Put } N = n+h$$

$$\begin{aligned}
&= \frac{8n}{9n-n-h} - \frac{n+n+h}{4n} \\
&= \frac{8n}{8n-h} - \frac{2n+h}{4n} \\
&= \frac{\cancel{8n}}{\cancel{8n} \left( 1 - \frac{h}{8n} \right)} - \frac{2n+h}{4n} \\
&= \left( 1 - \frac{h}{8n} \right)^{-1} - \frac{2n}{4n} - \frac{h}{4n} \\
&= 1 + \frac{h}{8n} - \frac{1}{2} - \frac{h}{4n} \text{ by neglecting } h^2, h^3, \dots \\
&\approx \frac{1}{2} - \frac{h}{8n}
\end{aligned}$$

$$\text{Hence L.H.S} \approx \text{R.H.S}$$



**Q.9 Identify the following series as binomial expansion and find the sum in each case.**

(i)  $1 - \frac{1}{2}\left(\frac{1}{4}\right) + \frac{1.3}{2!4}\left(\frac{1}{4}\right)^2 - \frac{1.3.5}{3!8}\left(\frac{1}{4}\right) + \dots$

**Solution:**

$$\text{Let } (1+x)^n = 1 - \frac{1}{2}\left(\frac{1}{4}\right) + \frac{1.3}{2!4}\left(\frac{1}{4}\right)^2 - \frac{1.3.5}{3!8}\left(\frac{1}{4}\right) + \dots \quad (\text{I})$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (\text{II})$$

Comparing (I) and (II)

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $nx = \frac{-1}{2}\left(\frac{1}{4}\right)$ $nx = -\frac{1}{8}$ $x = -\frac{1}{8n} \quad (\text{i})$ | $\frac{n(n-1)x^2}{2!} = \frac{1.3}{2 \times 4}\left(\frac{1}{4}\right)^2$ $n(n-1)x^2 = \frac{3}{4} \times \frac{1}{16}$ $n(n-1)\left(-\frac{1}{8n}\right)^2 = \frac{3}{64} \quad \text{using (i)}$ $n(n-1) \times \frac{1}{64n^2} = \frac{3}{64}$ $\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$ $n = -\frac{1}{2} \text{ Put in (i)}$ |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$$x = \frac{-1}{8\left(-\frac{1}{2}\right)} \Rightarrow x = \frac{1}{4}$$

$$\text{Now, the sum of the given series} = (1+x)^n = \left(1 + \frac{1}{4}\right)^{-\frac{1}{2}} = \left(\frac{5}{4}\right)^{-\frac{1}{2}} = \left(\frac{4}{5}\right)^{\frac{1}{2}} = \sqrt{\frac{4}{5}}$$

(ii)  $1 - \frac{1}{2}\left(\frac{1}{2}\right) + \frac{1.3}{2.4}\left(\frac{1}{2}\right)^2 - \frac{1.3.5}{2.4.6}\left(\frac{1}{2}\right)^3 + \dots$

**Solution:**

$$\text{Let } (1+x)^n = 1 - \frac{1}{2}\left(\frac{1}{2}\right) + \frac{1.3}{2.4}\left(\frac{1}{2}\right)^2 - \frac{1.3.5}{2.4.6}\left(\frac{1}{2}\right)^3 + \dots \quad (\text{I})$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (\text{II})$$

Comparing (I) and (II)

|                                                                 |                                                                                                                        |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| $nx = \frac{-1}{2}\left(\frac{1}{2}\right)$ $nx = -\frac{1}{4}$ | $\frac{n(n-1)x^2}{2!} = \frac{1.3}{2 \times 4}\left(\frac{1}{2}\right)^2$ $n(n-1)x^2 = \frac{3}{4} \times \frac{1}{4}$ |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|

$$x = -\frac{1}{4n} \quad (i)$$

$$n(n-1)\left(-\frac{1}{4n}\right)^2 = \frac{3}{16} \quad \text{using (i)}$$

$$n(n-1) \times \frac{1}{16n^2} = \frac{3}{16}$$

$$\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$$

$$n = \frac{-1}{2} \text{ Put in (i)}$$

$$x = \frac{-1}{4\left(-\frac{1}{2}\right)} \Rightarrow x = \frac{1}{2}$$

$$\text{Now, the sum of the given series} = (1+x)^n = \left(1+\frac{1}{2}\right)^{\frac{1}{2}} = \left(\frac{3}{2}\right)^{\frac{1}{2}} = \left(\frac{2}{3}\right)^{\frac{1}{2}} = \sqrt{\frac{2}{3}}$$

$$(iii) \quad 1 + \frac{3}{4} + \frac{3.5}{4.8} + \frac{3.5.7}{4.8.12} + \dots$$

**Solution:**

$$\text{Let } (1+x)^n = 1 + \frac{3}{4} + \frac{3.5}{4.8} + \frac{3.5.7}{4.8.12} + \dots \quad (I)$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (II)$$

Comparing (I) and (II)

$$\begin{aligned} nx &= \frac{3}{4} \\ x &= \frac{3}{4n} \end{aligned} \quad (i)$$

$$\frac{n(n-1)x^2}{2!} = \frac{3.5}{4.8}$$

$$n(n-1)x^2 = \frac{3.5}{2.8}$$

$$n(n-1)\left(\frac{3}{4n}\right)^2 = \frac{15}{16} \quad \text{using (i)}$$

$$n(n-1) \times \frac{9}{16n^2} = \frac{15}{16}$$

$$\frac{n-1}{n} = \frac{15}{9} \Rightarrow 3n-3 = 5n \Rightarrow 2n = -3$$

$$n = \frac{-3}{2} \text{ Put in (i)}$$

$$x = \frac{3}{4\left(-\frac{3}{2}\right)} \Rightarrow x = \frac{-1}{2}$$

Now, the sum of the given series

$$= (1+x)^n = \left(1 - \frac{1}{2}\right)^{-\frac{3}{2}} = \left(\frac{1}{2}\right)^{-\frac{3}{2}} = (2)^{\frac{3}{2}} = (2^3)^{\frac{1}{2}} = \sqrt{8} = 2\sqrt{2}$$

$$(iv) \quad 1 - \frac{1}{2}\left(\frac{1}{3}\right) + \frac{1.3}{2.4}\left(\frac{1}{3}\right)^2 - \frac{1.3.5}{2.4.6}\left(\frac{1}{3}\right)^3 + \dots$$

**Solution:**

$$\text{Let } (1+x)^n = 1 - \frac{1}{2}\left(\frac{1}{3}\right) + \frac{1.3}{2.4}\left(\frac{1}{3}\right)^2 - \frac{1.3.5}{2.4.6}\left(\frac{1}{3}\right)^3 + \dots \quad (I)$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (II)$$

Comparing (I) and (II)

|                                                                                               |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $nx = \frac{-1}{2}\left(\frac{1}{3}\right)$ $nx = -\frac{1}{6}$ $x = -\frac{1}{6n} \quad (i)$ | $\frac{n(n-1)x^2}{2!} = \frac{1.3}{2.4}\left(\frac{1}{3}\right)^2$ $n(n-1)x^2 = \frac{3}{4} \times \frac{1}{9}$ $n(n-1)\left(-\frac{1}{6n}\right)^2 = \frac{3}{36} \quad \text{using (i)}$ $n(n-1) \times \frac{1}{36n^2} = \frac{3}{36}$ $\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$ $n = \frac{-1}{2} \text{ Put in (i)}$ |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$$x = \frac{-1}{6\left(-\frac{1}{2}\right)} \Rightarrow x = \frac{1}{3}$$

$$\text{Now, the sum of the given series} = (1+x)^n = \left(1 + \frac{1}{3}\right)^{-\frac{1}{2}} = \left(\frac{4}{3}\right)^{-\frac{1}{2}} = \left(\frac{3}{4}\right)^{\frac{1}{2}} = \frac{\sqrt{3}}{2}$$

**Q.10** Use binomial theorem to show that  $1 + \frac{1}{4} + \frac{1.3}{4.8} + \frac{1.3.5}{4.8.12} + \dots = \sqrt{2}$

$$\text{Proof: Let } (1+x)^n = 1 + \frac{1}{4} + \frac{1.3}{4.8} + \frac{1.3.5}{4.8.12} + \dots \quad (I)$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (II)$$

Comparing (I) and (II)

|                    |                                          |
|--------------------|------------------------------------------|
| $nx = \frac{1}{4}$ | $\frac{n(n-1)x^2}{2!} = \frac{1.3}{4.8}$ |
|--------------------|------------------------------------------|

$$x = \frac{1}{4n}$$

$$(i) \quad n(n-1)x^2 = \frac{1.3}{2.8}$$

$$n(n-1)\left(\frac{1}{4n}\right)^2 = \frac{3}{16} \quad \text{using (i)}$$

$$n(n-1) \times \frac{1}{16n^2} = \frac{3}{16}$$

$$\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$$

$$n = \frac{-1}{2} \text{ Put in (i)}$$

$$x = \frac{1}{4\left(-\frac{1}{2}\right)} \Rightarrow x = \frac{-1}{2}$$

$$\text{Now, the sum of the given series} = (1+x)^n = \left(1 - \frac{1}{2}\right)^{-\frac{1}{2}} = \left(\frac{1}{2}\right)^{-\frac{1}{2}} = (2)^{\frac{1}{2}} = \sqrt{2}$$

Hence the proof

**Q.11** If  $y = \frac{1}{3} + \frac{1.3}{2!}\left(\frac{1}{3}\right)^2 + \frac{1.3.5}{3!}\left(\frac{1}{3}\right)^3 + \dots$

**Then prove that**  $y^2 + 2y - 2 = 0$

**Proof:** Given that

$$y = \frac{1}{3} + \frac{1.3}{2!}\left(\frac{1}{3}\right)^2 + \frac{1.3.5}{3!}\left(\frac{1}{3}\right)^3 + \dots$$

Adding 1 on both sides of given series of make it binomial series

$$1+y = 1 + \frac{1}{3} + \frac{1.3}{2!}\left(\frac{1}{3}\right)^2 + \frac{1.3.5}{3!}\left(\frac{1}{3}\right)^3 + \dots \quad (I)$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (II)$$

Comparing (I) and (II)

$$nx = \frac{1}{3}$$

$$x = \frac{1}{3n}$$

(i)

$$\frac{n(n-1)x^2}{2!} = \frac{1.3}{2!}\left(\frac{1}{3}\right)^2$$

$$n(n-1)x^2 = \frac{3}{9}$$

$$n(n-1)\left(\frac{1}{3n}\right)^2 = \frac{3}{9} \quad \text{using (i)}$$

$$n(n-1) \times \frac{1}{9n^2} = \frac{3}{9}$$

$$\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$$

$$n = \frac{-1}{2} \text{ Put in (i)}$$

$$x = \frac{1}{3\left(-\frac{1}{2}\right)} \Rightarrow x = \frac{-2}{3}$$

From (I) and (II)

$$1 + y = (1 + x)^n$$

$$1 + y = \left(1 - \frac{2}{3}\right)^{\frac{-1}{2}}$$

$$1 + y = \left(\frac{1}{3}\right)^{\frac{-1}{2}}$$

$$1 + y = \sqrt{3}$$

Squaring on both sides

$$1 + y^2 + 2y = 3$$

$$y^2 + 2y - 2 = 0$$

Which is required to prove.

**Q.12** If  $2y = \frac{1}{2^2} + \frac{1.3}{2!} \cdot \frac{1}{2^4} + \frac{1.3.5}{3!} \cdot \frac{1}{2^6} + \dots$

**Then Prove that  $4y^2 + 4y - 1 = 0$**

**Proof:** Given that

$$2y = \frac{1}{2^2} + \frac{1.3}{2!} \cdot \frac{1}{2^4} + \frac{1.3.5}{3!} \cdot \frac{1}{2^6} + \dots$$

Adding 1 on both sides to make it binomial series, we get ;

$$1 + 2y = 1 + \frac{1}{2^2} + \frac{1.3}{2!} \times \frac{1}{2^4} + \frac{1.3.5}{3!} \times \frac{1}{2^6} + \dots \quad (I)$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots \quad (II)$$

Comparing (I) and (II)

$$\begin{aligned} nx &= \frac{1}{2^2} \\ x &= \frac{1}{4n} \end{aligned} \quad (i)$$

$$\frac{n(n-1)x^2}{2!} = \frac{1.3}{2!} \cdot \frac{1}{2^4}$$

$$n(n-1)x^2 = \frac{3}{16}$$

$$n(n-1) \left( \frac{1}{4n} \right)^2 = \frac{3}{16} \quad \text{using (i)}$$

$$n(n-1) \times \frac{1}{16n^2} = \frac{3}{16}$$

$$\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$$

$$n = \frac{-1}{2} \quad \text{Put in (i)}$$

$$x = \frac{1}{4 \left( -\frac{1}{2} \right)} \Rightarrow x = \frac{-1}{2}$$

From (I) and (II)

$$1 + 2y = (1+x)^n$$

$$1 + 2y = \left( 1 - \frac{1}{2} \right)^{-1}$$

$$1 + 2y = \left( \frac{1}{2} \right)^{-1}$$

$$1 + 2y = \sqrt{2}$$

Squaring on both sides

$$1 + 4y^2 + 4y = 2$$

$$4y^2 + 4y - 1 = 0$$

Which is required to prove.

**Q.13** If  $y = \frac{2}{5} + \frac{1.3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1.3.5}{3!} \left(\frac{2}{5}\right)^3 + \dots$

**Then prove that  $y^2 + 2y - 4 = 0$**

**Proof:** Give that

$$y = \frac{2}{5} + \frac{1.3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1.3.5}{3!} \left(\frac{2}{5}\right)^3 + \dots$$

Adding 1 on both sides of given series to make it binomial series.

$$\text{So, } 1 + y = 1 + \frac{2}{5} + \frac{1.3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1.3.5}{3!} \left(\frac{2}{5}\right)^3 + \dots \quad (\text{I})$$

$$\text{As we know } (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots \quad (\text{II})$$

Comparing (I) and (II)

$$nx = \frac{2}{5}$$

$$x = \frac{2}{5n}$$

(i)

$$\frac{n(n-1)x^2}{2!} = \frac{1.3}{2!} \left(\frac{2}{5}\right)^2$$

$$n(n-1)x^2 = \frac{12}{25}$$

$$n(n-1) \left(\frac{2}{5n}\right)^2 = \frac{12}{25} \quad \text{using (i)}$$

$$n(n-1) \times \frac{4}{25n^2} = \frac{12}{25}$$

$$\frac{n-1}{n} = 3 \Rightarrow n-1 = 3n \Rightarrow 2n = -1$$

$$n = \frac{-1}{2} \quad \text{Put in (i)}$$

$$x = \frac{2}{5 \left(-\frac{1}{2}\right)} \Rightarrow x = \frac{-4}{5}$$

From (I) and (II)

$$1 + y = (1+x)^n$$

$$1 + y = \left(1 - \frac{4}{5}\right)^{\frac{-1}{2}}$$

$$1 + y = \left(\frac{1}{5}\right)^{\frac{-1}{2}}$$

$$1 + y = \sqrt{5}$$

Squaring on both sides

$$1 + y^2 + 2y = 5$$

$$y^2 + 2y - 4 = 0$$

Which is required to prove.